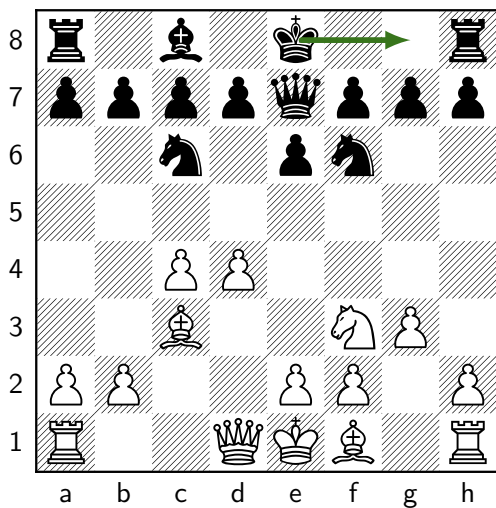


Melanie-BOTS (CCRL engines) — Bogo-Indian

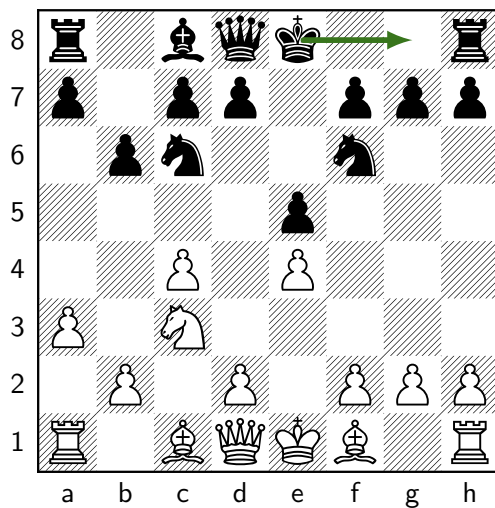
Melanie (Deep_Vision)

Abstract

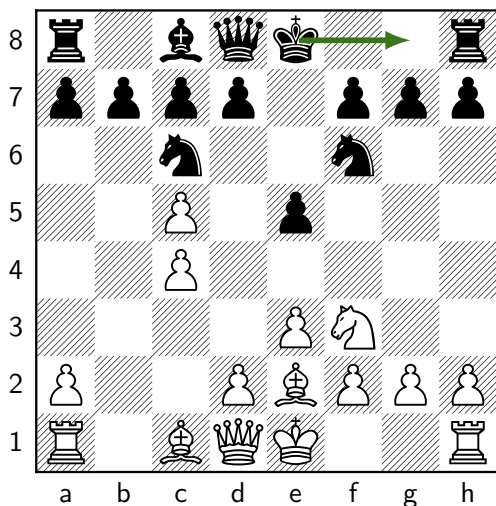
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



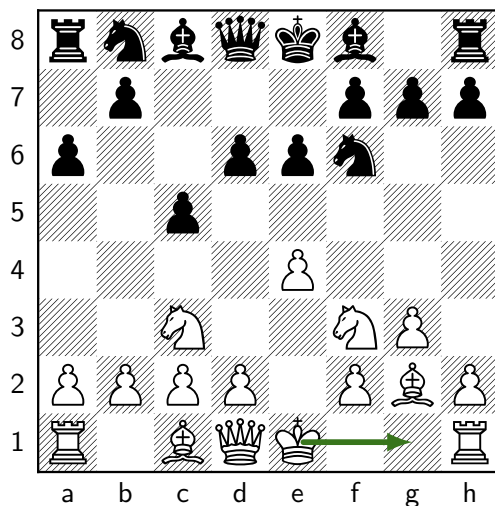
(a)



(b)

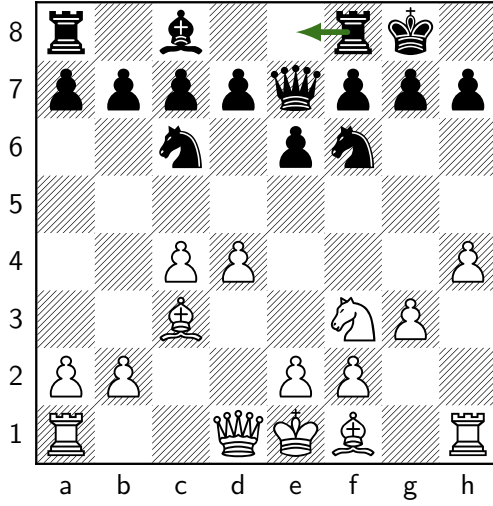


(c)

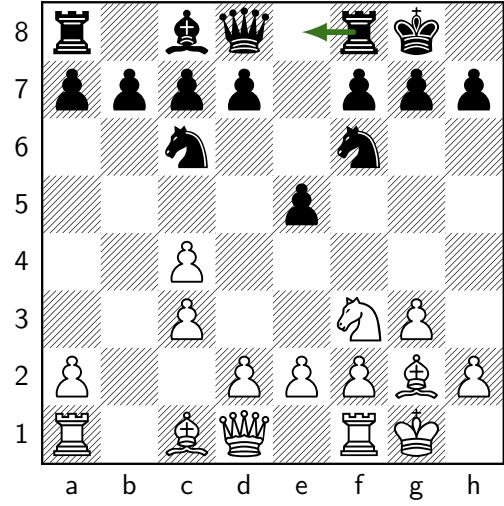


(d)

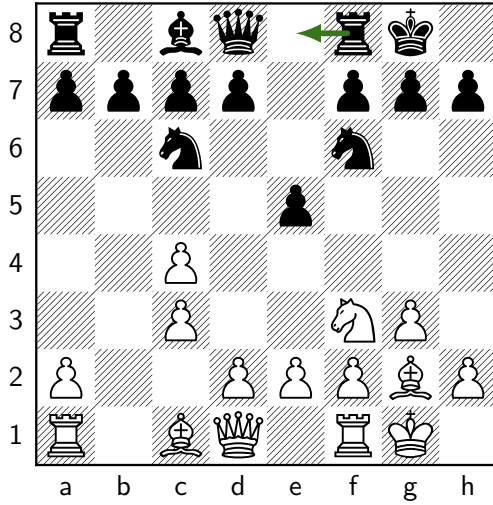
Figure 1: **Move 7, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0003#351` (ply 13) [structural distance $d=47$, QAP obj=29]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pc7* as their *Ra8*, our *Pf7* as their *Bc8*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccl1_top_p0001#666` (ply 15) [structural distance $d=42$, QAP obj=25]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf6* as their *Ra8*, our *pa2* as their *Bc8*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccl1_top_p0000#937` (ply 10) [structural distance $d=53$, QAP obj=24]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pb7* as their *Ra1*, our *Nf6* as their *Bc1*; transferred move e8g8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to 0-0 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



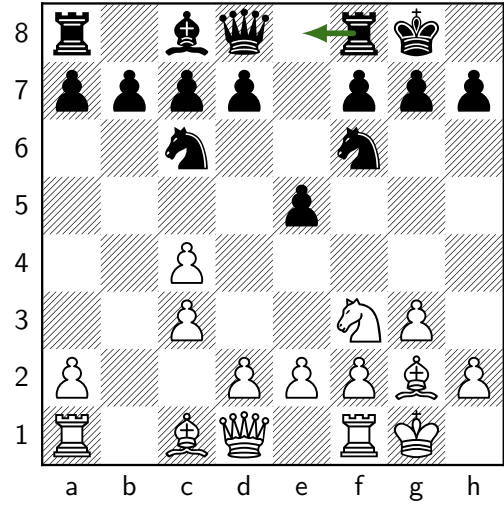
(a)



(b)

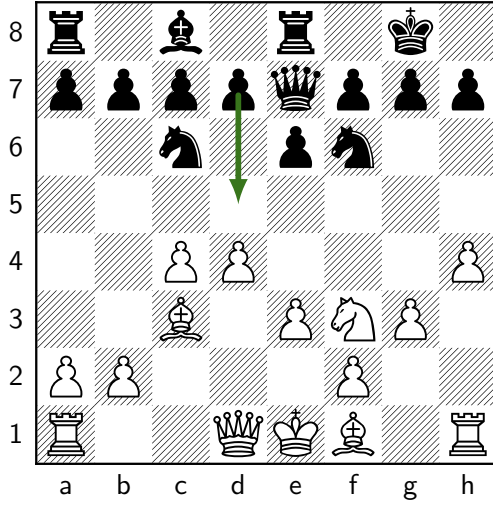


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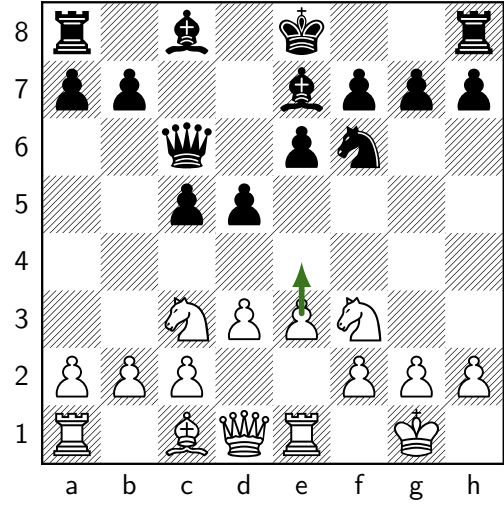


(d)

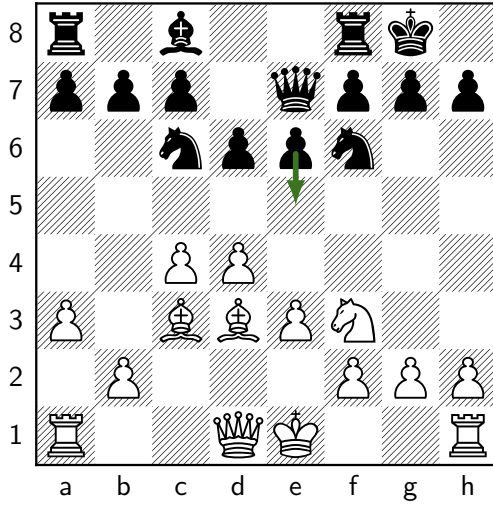
Figure 2: **Move 8, Black: Re8.** (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1270` (ply 13) [structural distance $d=45$, QAP obj=25]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pg3* as their *Ra8*, our *bc3* as their *Qd8*; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1271` (ply 13) [structural distance $d=45$, QAP obj=25]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pg3* as their *Ra8*, our *bc3* as their *Qd8*; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3522` (ply 13) [structural distance $d=45$, QAP obj=25]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pg3* as their *Ra8*, our *bc3* as their *Qd8*; transferred move f8e8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Re8 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



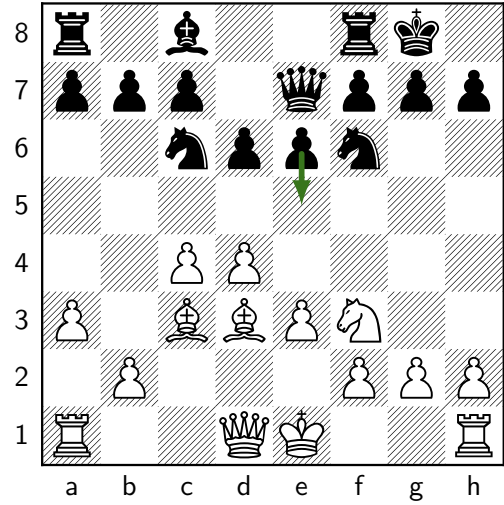
(a)



(b)

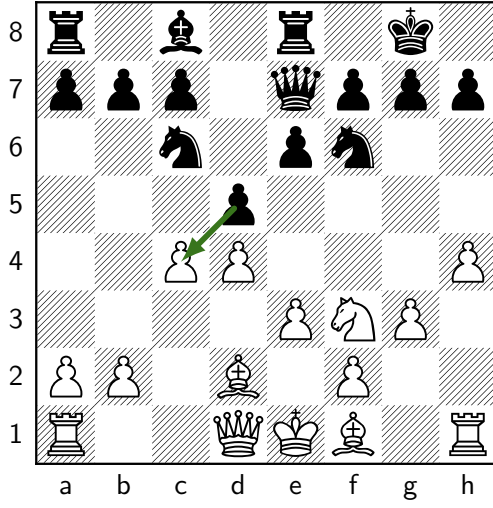


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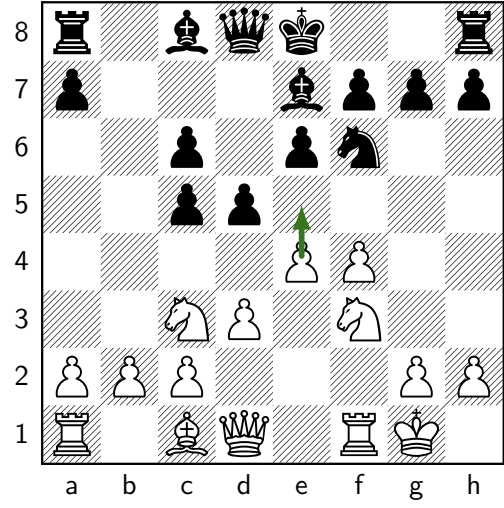


(d)

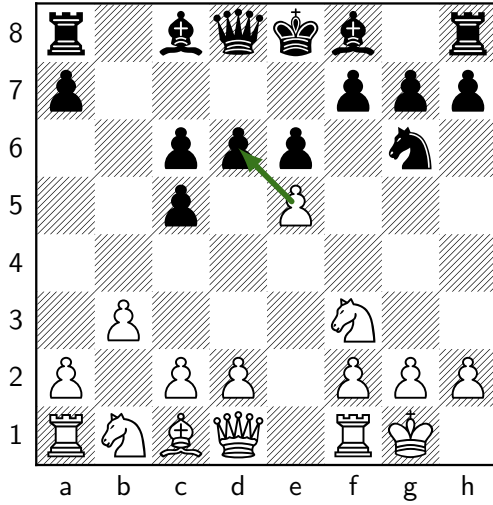
Figure 3: **Move 9, Black: d5.** (a) Our position; Melanie plays d5 (d7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#134` (ply 16) [structural distance $d=48$, QAP obj=30]: the master played e4; under μ our $Pd2$ plays the role of that game’s $Pe3$; cross-type roles: our $pb2$ as their $Ra1$, our $Pa7$ as their $Bc1$; transferred move d7d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3433` (ply 17) [structural distance $d=47$, QAP obj=34]: the master played e5; under μ our $Nf3$ plays the role of that game’s $Pe3$ (a **cross-type** correspondence); cross-type roles: our $pg3$ as their $Bc8$, our $Bc8$ as their $Pa7$; transferred move f6d5. (d) Retrieved from `ccr1_top_p0001#3434` (ply 17) [structural distance $d=47$, QAP obj=34]: the master played e5; under μ our $Nf3$ plays the role of that game’s $Pe3$ (a **cross-type** correspondence); cross-type roles: our $pg3$ as their $Bc8$, our $Bc8$ as their $Pa7$; transferred move f6d5. Synthesis: 1 retrieved analog transferred to d5 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



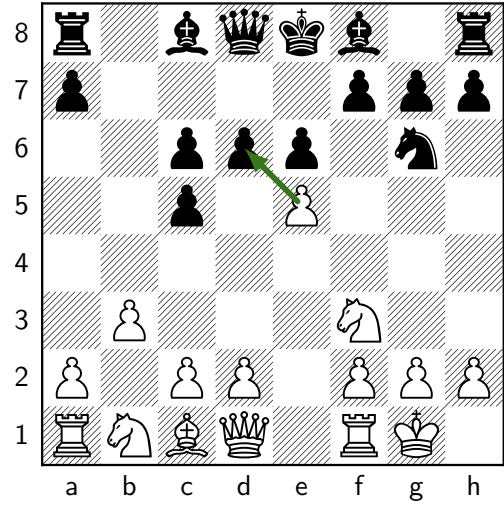
(a)



(b)

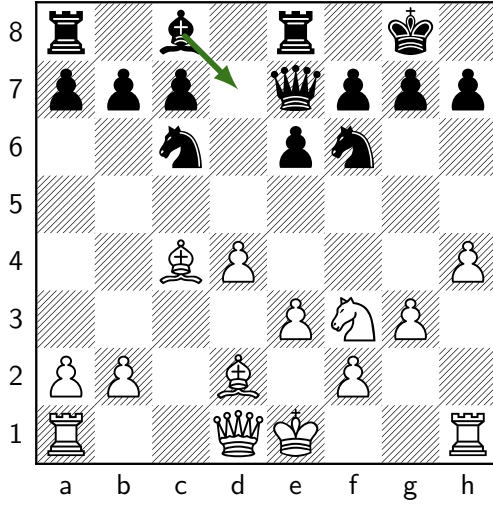


(c)

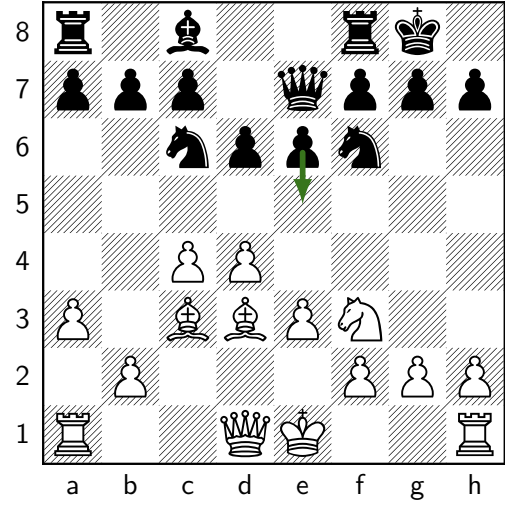


(d)

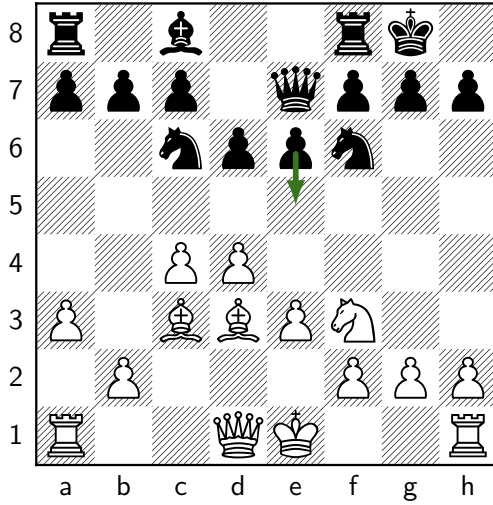
Figure 4: **Move 10, Black: dxc4.** (a) Our position; Melanie plays dxc4 (d5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2427` (ply 16) [structural distance $d=39$, QAP obj=58]: the master played e5; under μ our $Pd4$ plays the role of that game’s $Pe4$; cross-type roles: our $pd4$ as their $Ra1$, our $pf2$ as their $Bc1$; transferred move d5c4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#242` (ply 14) [structural distance $d=61$, QAP obj=54]: the master played exd6; under μ our $Pd4$ plays the role of that game’s $Pe5$; cross-type roles: our $Qe7$ as their $Nb1$, our $Pb7$ as their $Bc1$; transferred move d5c4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#243` (ply 14) [structural distance $d=61$, QAP obj=54]: the master played exd6; under μ our $Pd4$ plays the role of that game’s $Pe5$; cross-type roles: our $Qe7$ as their $Nb1$, our $Pb7$ as their $Bc1$; transferred move d5c4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to dxc4 (summed QAP objective 165, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



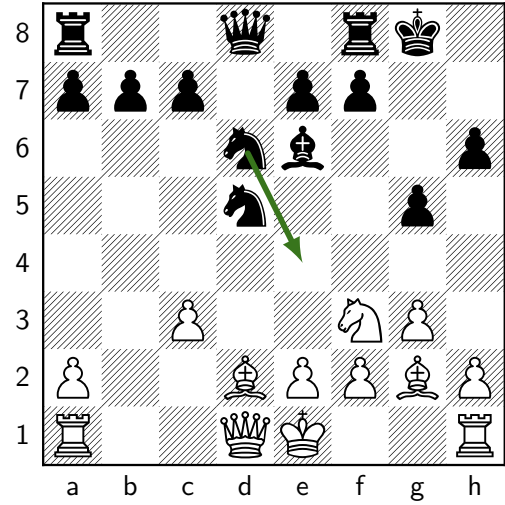
(a)



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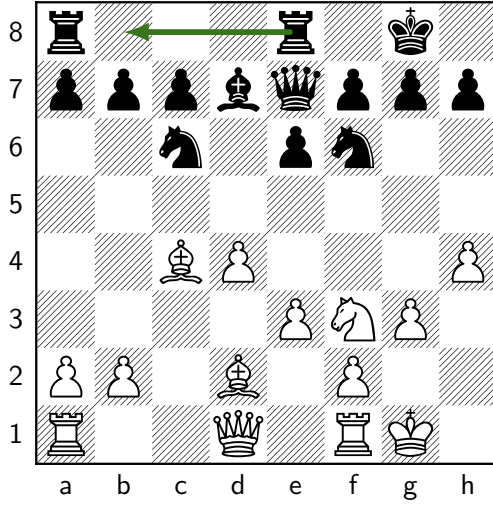


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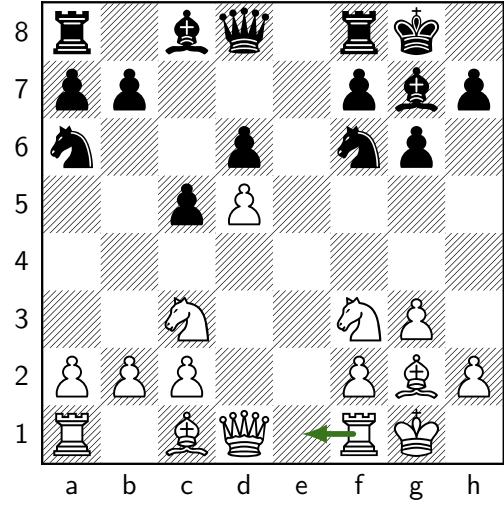


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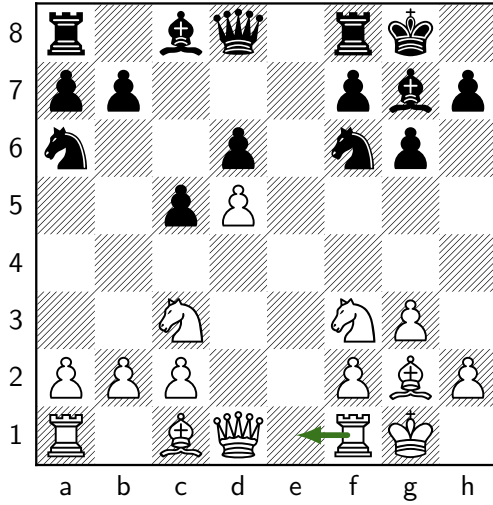
Figure 5: **Move 11, Black: Bd7.** (a) Our position; Melanie plays Bd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3433` (ply 17) [structural distance $d=52$, QAP obj=36]: the master played e5; under μ our *Bc1* plays the role of that game’s *Pe3* (a **cross-type** correspondence); cross-type roles: our *bd2* as their *Ra8*, our *Ra8* as their *Bc8*; transferred move c8d7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3434` (ply 17) [structural distance $d=52$, QAP obj=36]: the master played e5; under μ our *Bc1* plays the role of that game’s *Pe3* (a **cross-type** correspondence); cross-type roles: our *bd2* as their *Ra8*, our *Ra8* as their *Bc8*; transferred move c8d7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#369` (ply 31) [structural distance $d=45$, QAP obj=52]: the master played Ne4; under μ our *Nf3* plays the role of that game’s *Nd3*; cross-type roles: our *Pa7* as their *Qd8*, our *Pf7* as their *Rf8*; transferred move f6e4. Synthesis: 2 retrieved analogs transferred to Bd7 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



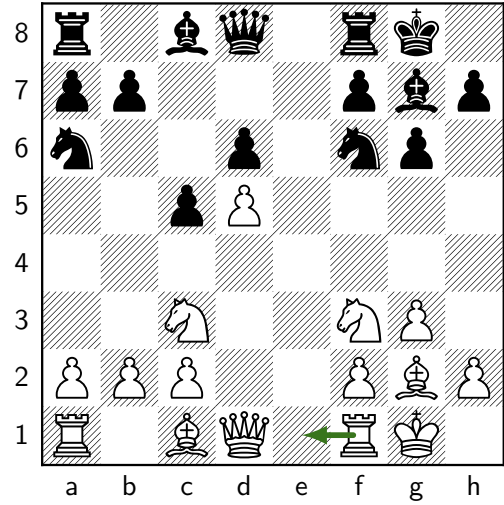
(a)



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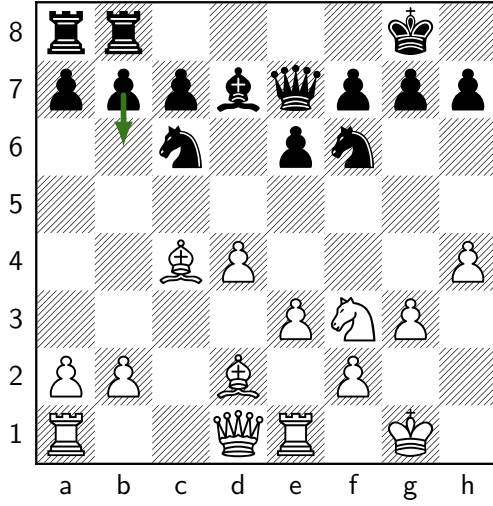


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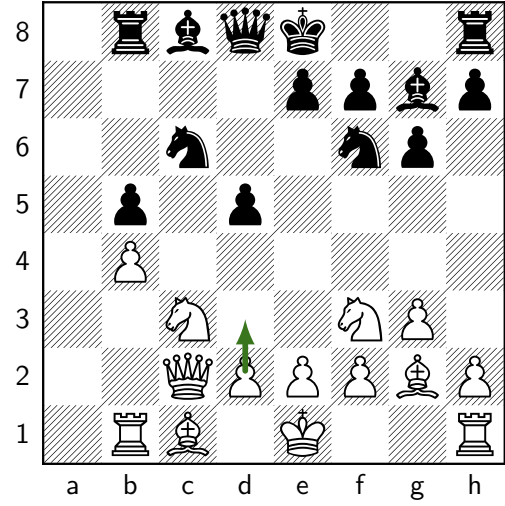


(d)

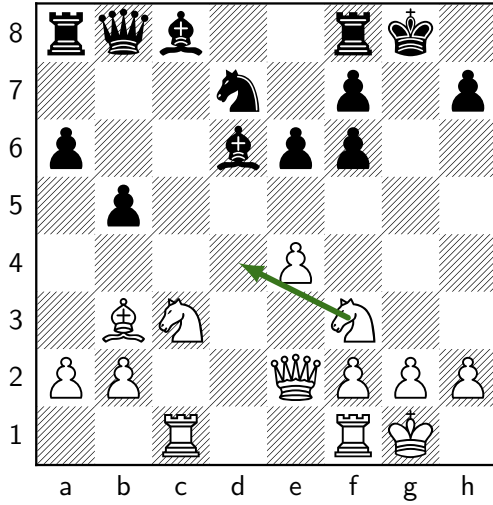
Figure 6: **Move 12, Black: Reb8.** (a) Our position; Melanie plays Reb8 (e8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0001#2113` (ply 18) [structural distance $d=68$, QAP obj=25]: the master played Re1; under μ our Re1 plays the role of that game’s Rf1; cross-type roles: our ph4 as their Ra1, our pd4 as their Bc1; transferred move e8b8 (**endorses this move**). (c) Retrieved from `ccr1.top_p0001#2114` (ply 18) [structural distance $d=68$, QAP obj=25]: the master played Re1; under μ our Re1 plays the role of that game’s Rf1; cross-type roles: our ph4 as their Ra1, our pd4 as their Bc1; transferred move e8b8 (**endorses this move**). (d) Retrieved from `ccr1.top_p0003#2376` (ply 18) [structural distance $d=68$, QAP obj=25]: the master played Re1; under μ our Re1 plays the role of that game’s Rf1; cross-type roles: our ph4 as their Ra1, our pd4 as their Bc1; transferred move e8b8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Reb8 (summed QAP objective 98, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



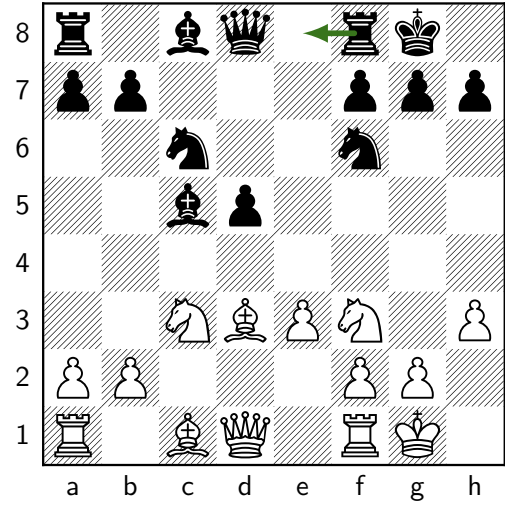
(a)



(b)

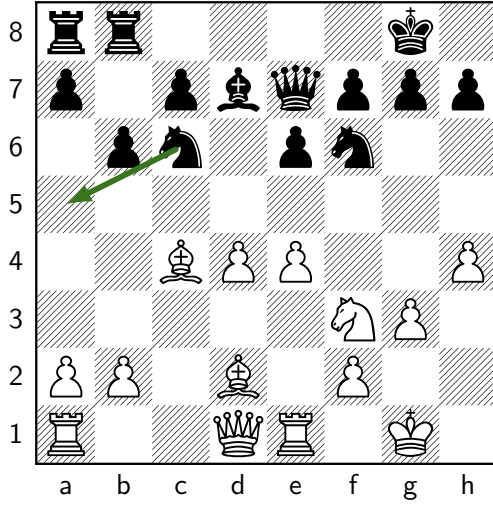


(c)

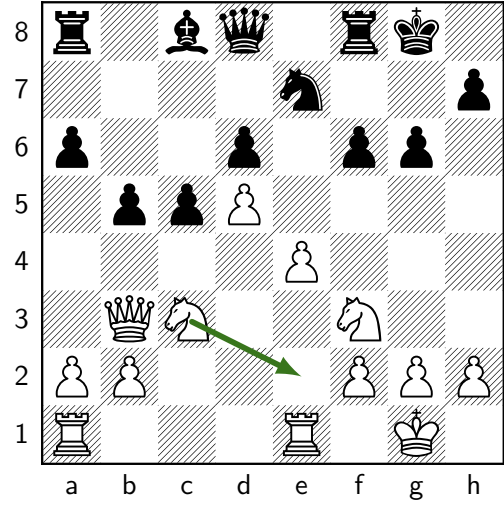


(d)

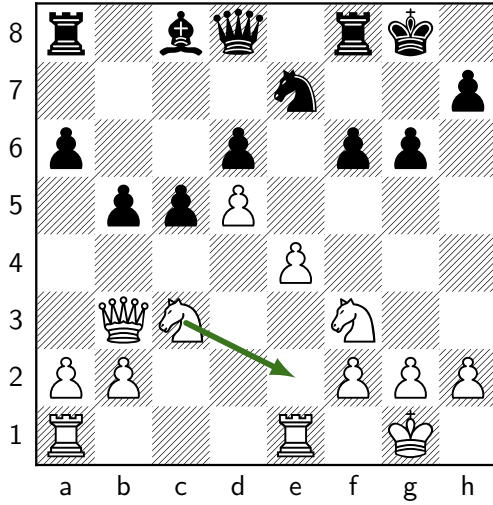
Figure 7: **Move 13, Black: b6**. (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1795` (ply 22) [structural distance $d=68$, QAP obj=64]: the master played d3; under μ our $Pb2$ plays the role of that game’s $Pd2$; cross-type roles: our $pa2$ as their $Bc1$, our $Nf6$ as their $Rh1$; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#88` (ply 28) [structural distance $d=77$, QAP obj=52]: the master played Nd4; under μ our $Nf3$ plays the role of that game’s $Nf3$; cross-type roles: our $Pc7$ as their $Rc1$, our $Qe7$ as their $Rf1$; transferred move f6e4. (d) Retrieved from `ccrl_top_p0003#1099` (ply 19) [structural distance $d=80$, QAP obj=47]: the master played Re8; under μ our $Nf3$ plays the role of that game’s $Rf1$ (a **cross-type** correspondence); cross-type roles: our $Pc7$ as their $Ra8$, our $Nc6$ as their $Bc8$; transferred move f6e8. Synthesis: 1 retrieved analog transferred to b6 (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



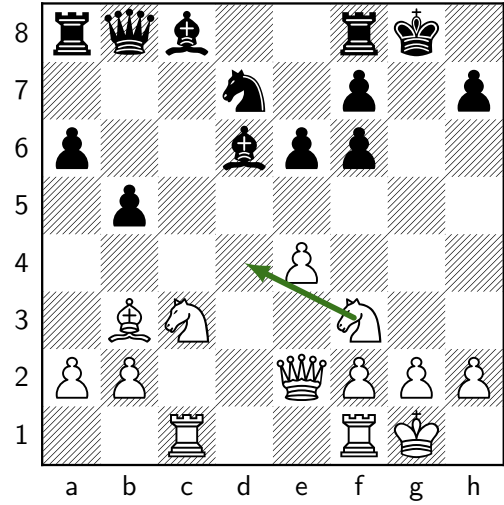
(a)



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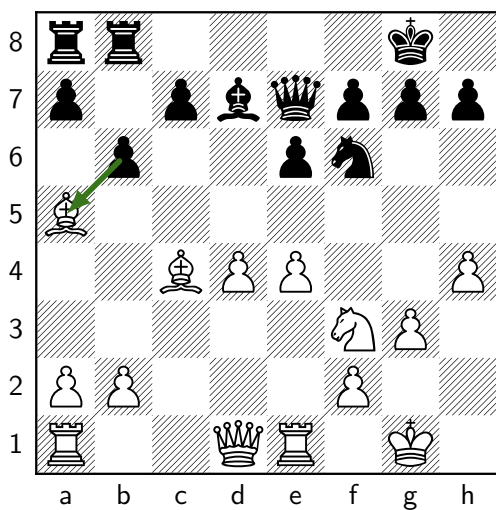


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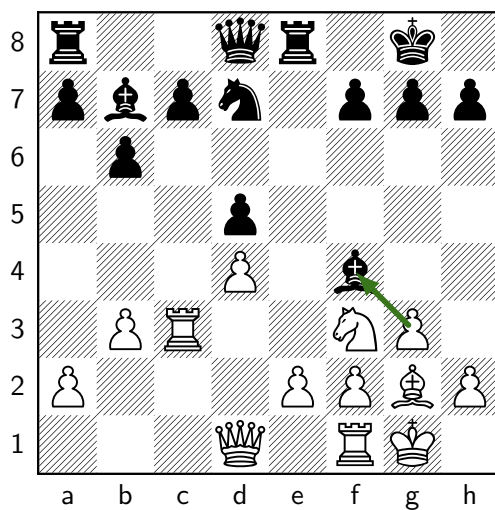


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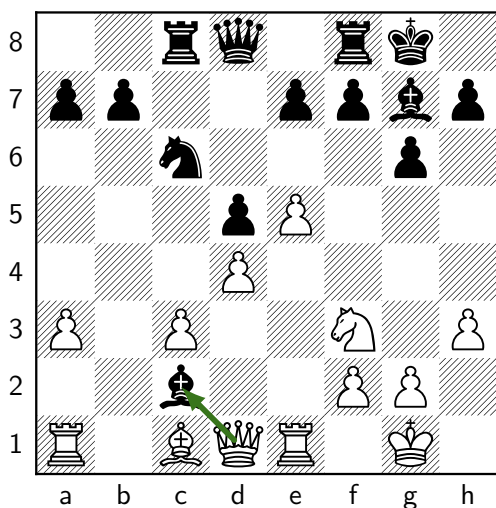
Figure 8: **Move 14, Black: Na5**. (a) Our position; Melanie plays Na5 (c6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1133` (ply 30) [structural distance $d=79$, QAP obj=57]: the master played Ne2; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our Ph7 as their Re1, our Bd7 as their Pb2; transferred move c6a5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2582` (ply 30) [structural distance $d=79$, QAP obj=57]: the master played Ne2; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our Ph7 as their Re1, our Bd7 as their Pb2; transferred move c6a5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#88` (ply 28) [structural distance $d=76$, QAP obj=58]: the master played Nd4; under μ our Qe2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Rc1, our Ph7 as their Rf1; transferred move e7d6. Synthesis: 2 retrieved analogs transferred to Na5 (summed QAP objective 113, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



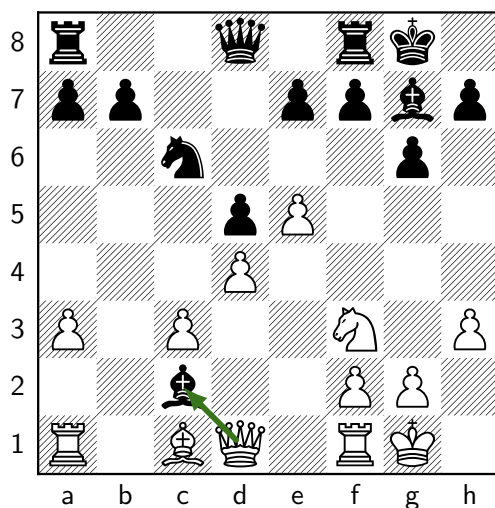
(a)



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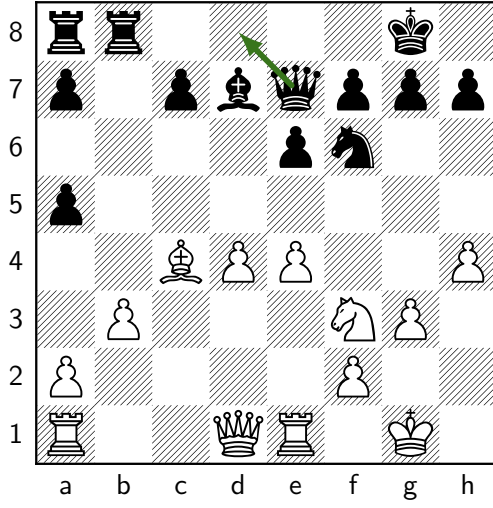


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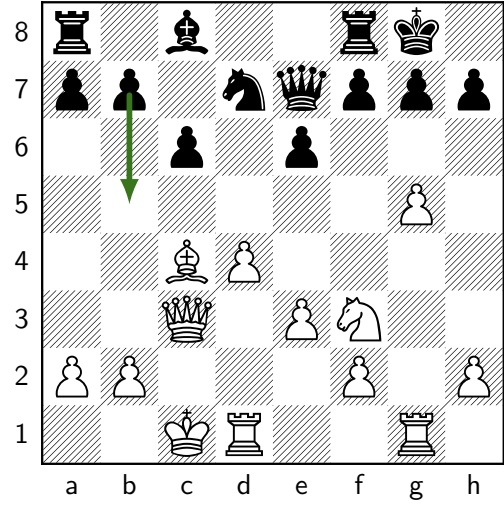


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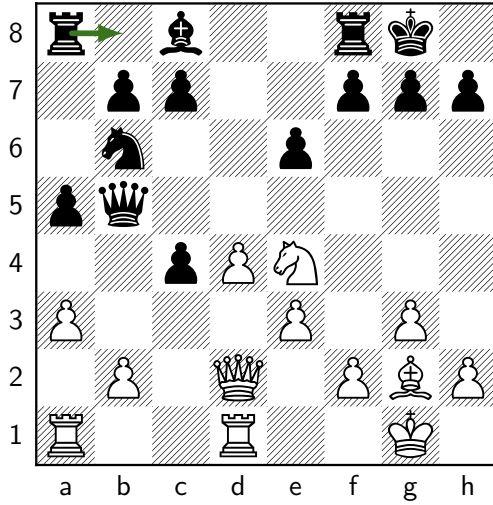
Figure 9: **Move 15, Black: bxa5**. (a) Our position; Melanie plays **bxa5** (**b6a5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1811` (ply 28) [structural distance $d=65$, QAP obj=30]: the master played **gxf4**; under μ our *Pb3* plays the role of that game’s *Pg3*; cross-type roles: our *bc4* as their *Qd1*, our *Nf6* as their *Rf1*; transferred move **b6a5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#493` (ply 28) [structural distance $d=61$, QAP obj=13]: the master played **Qxc2**; under μ our *Pb3* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Pe6* as their *Bc1*, our *Qe7* as their *Pc3*; transferred move **b6a5** (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2543` (ply 26) [structural distance $d=64$, QAP obj=12]: the master played **Qxc2**; under μ our *Pb3* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pf2* as their *Ra1*, our *Nf6* as their *Bc1*; transferred move **b6a5** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **bxa5** (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



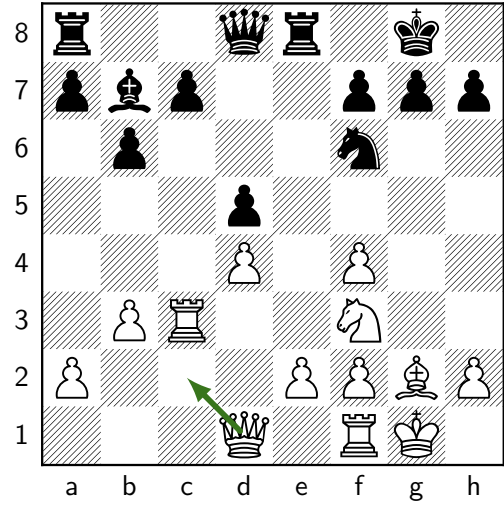
(a)



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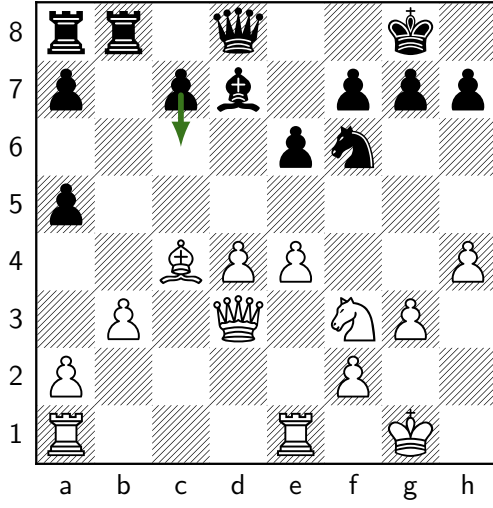


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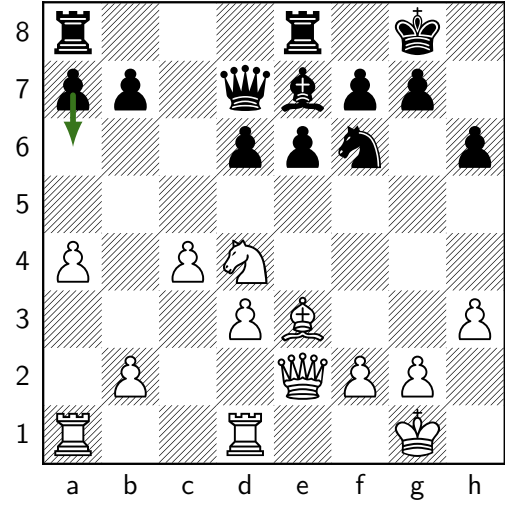


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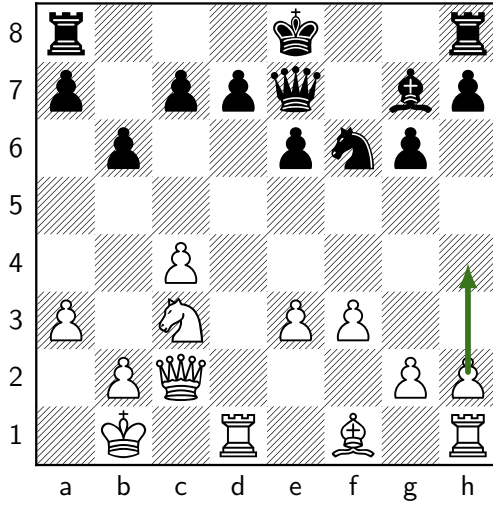
Figure 10: **Move 16, Black: Qd8.** (a) Our position; Melanie plays Qd8 (e7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3949` (ply 27) [structural distance $d=56$, QAP obj=43]: the master played b5; under μ our *Qe2* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *Ra8* as their *Bc8*, our *Ph7* as their *Rf8*; transferred move e7d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3213` (ply 29) [structural distance $d=62$, QAP obj=26]: the master played Rb8; under μ our *Qe2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Rb8* as their *Bc8*, our *qd1* as their *Rf8*; transferred move e7d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1811` (ply 30) [structural distance $d=57$, QAP obj=22]: the master played Qc2; under μ our *Qe2* plays the role of that game’s *Qd1*; cross-type roles: our *Ra8* as their *Pa2*, our *pf2* as their *Bg2*; transferred move e7d8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd8 (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



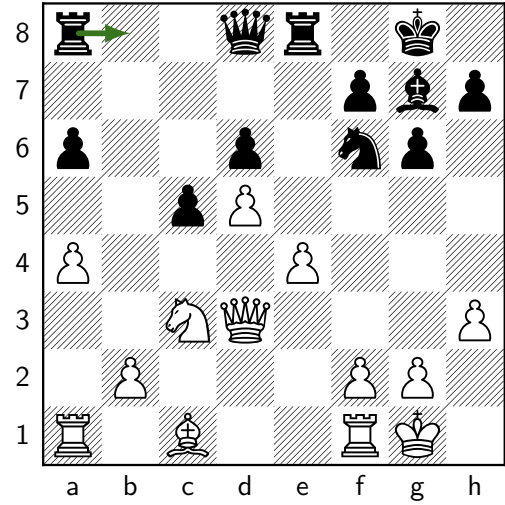
(a)



(b)

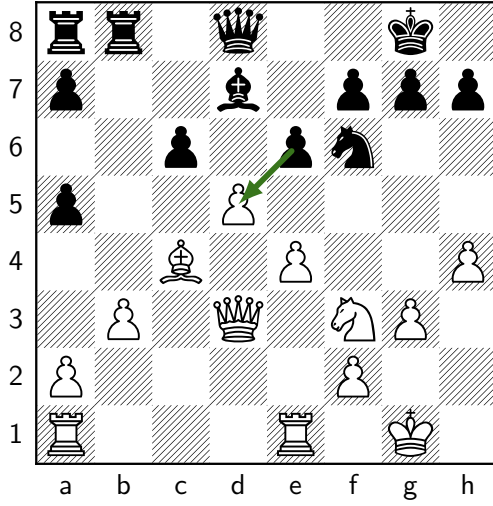


(c)

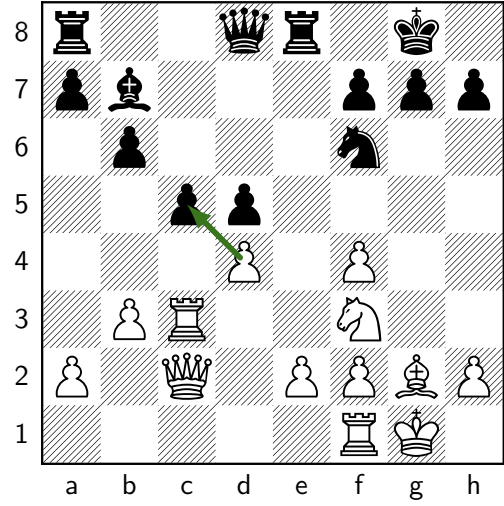


(d)

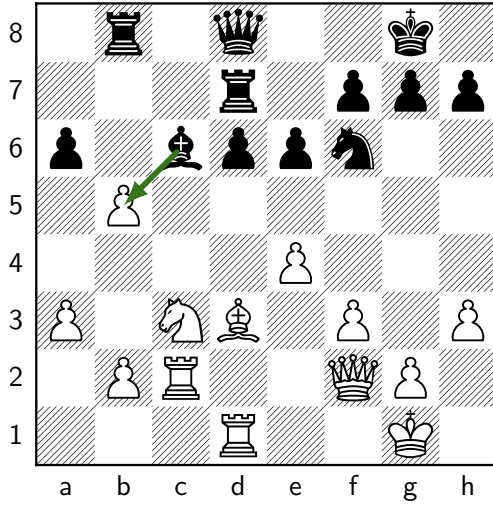
Figure 11: **Move 17, Black: c6.** (a) Our position; Melanie plays c6 (c7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3257` (ply 31) [structural distance $d=57$, QAP obj=45]: the master played a6; under μ our *Pc2* plays the role of that game’s *Pa2*; cross-type roles: our *pg3* as their *Ra8*, our *Ph7* as their *Re8*; transferred move c7c6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#637` (ply 26) [structural distance $d=71$, QAP obj=30]: the master played h4; under μ our *Pc2* plays the role of that game’s *Ph2*; cross-type roles: our *Nf6* as their *Rd1*, our *pg3* as their *Bf1*; transferred move c7c6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3879` (ply 35) [structural distance $d=70$, QAP obj=67]: the master played Rb8; under μ our *Bd2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Nf6* as their *Re8*, our *Pg7* as their *Bg7*; transferred move d7e8. Synthesis: 2 retrieved analogs transferred to c6 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



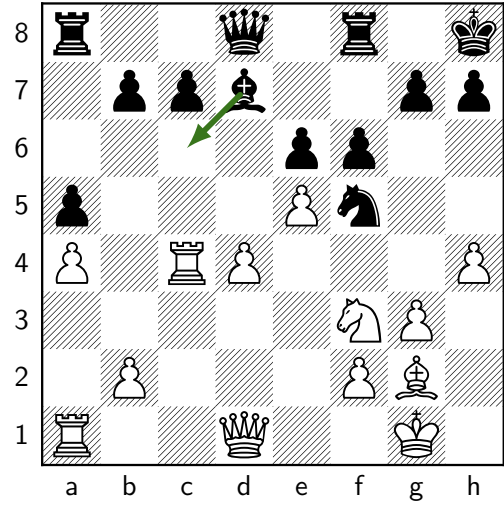
(a)



(b)

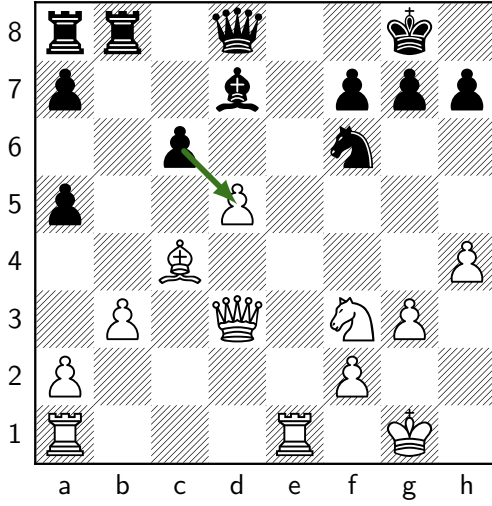


(c)

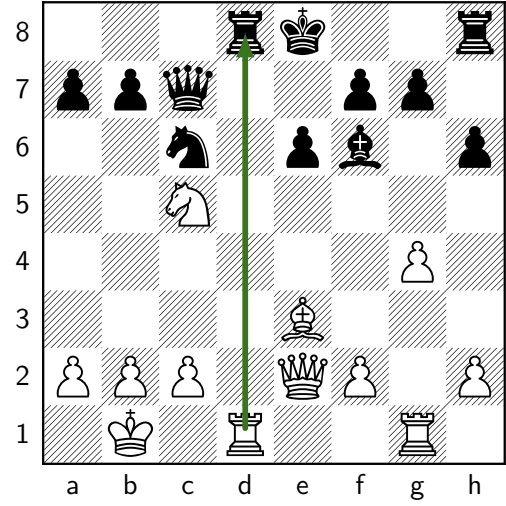


(d)

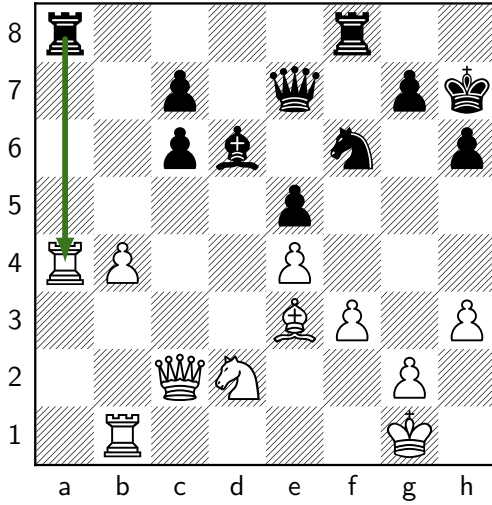
Figure 12: **Move 18, Black: exd5.** (a) Our position; Melanie plays **exd5** (**e6d5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1811` (ply 32) [structural distance $d=74$, QAP obj=73]: the master played **dx c5**; under μ our *Pe3* plays the role of that game’s *Pd4*; cross-type roles: our *Pf7* as their *Rf1*, our *Ra8* as their *Pa2*; transferred move **e6d5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#556` (ply 45) [structural distance $d=65$, QAP obj=52]: the master played **Bxb5**; under μ our *Pe3* plays the role of that game’s *Bc3* (a **cross-type** correspondence); cross-type roles: our *Nf6* as their *Rb8*, our *Bd7* as their *Qd8*; transferred move **e6d5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#509` (ply 35) [structural distance $d=78$, QAP obj=96]: the master played **Bc6**; under μ our *Pc3* plays the role of that game’s *Bd2* (a **cross-type** correspondence); cross-type roles: our *Pa7* as their *Ra8*, our *Pf7* as their *Qd8*; transferred move **c6c5**. Synthesis: 2 retrieved analogs transferred to **exd5** (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



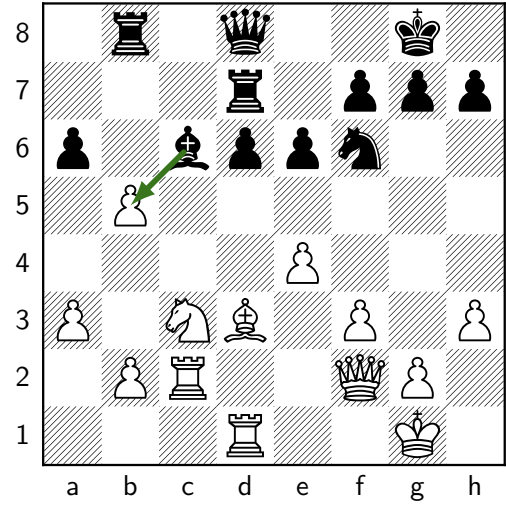
(a)



(b)

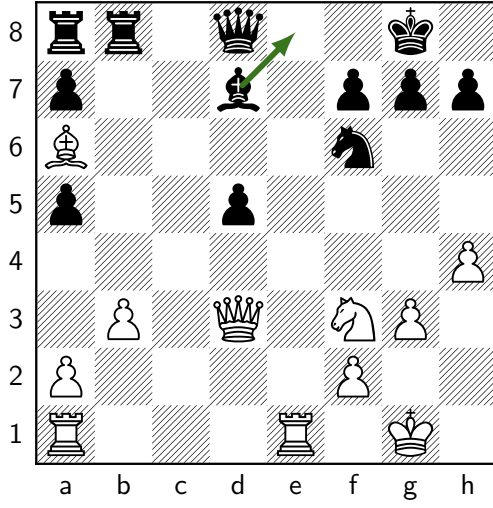


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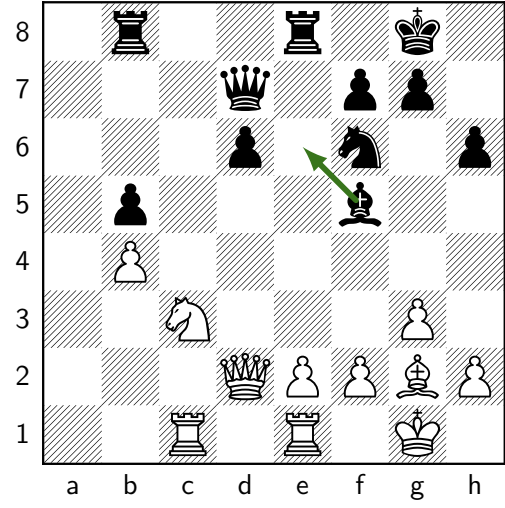


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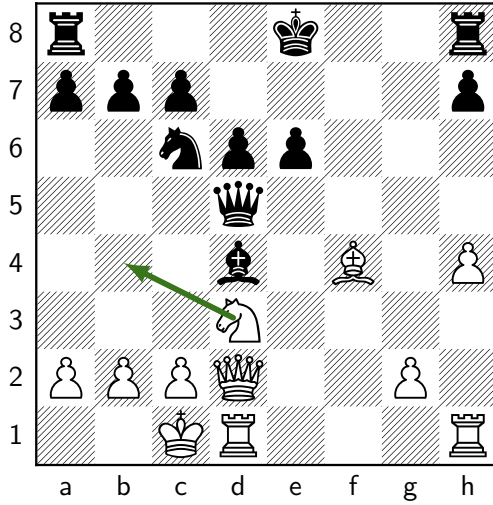
Figure 13: **Move 19, Black: cxd5.** (a) Our position; Melanie plays cxd5 (c6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#45` (ply 34) [structural distance $d=70$, QAP obj=48]: the master played $Rxd8+$; under μ our $Pc3$ plays the role of that game’s $Rd1$ (a **cross-type** correspondence); cross-type roles: our $Qd8$ as their $Rg1$, our $Bd7$ as their $Qe2$; transferred move c6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3694` (ply 41) [structural distance $d=69$, QAP obj=45]: the master played $Rxa4$; under μ our $Pc3$ plays the role of that game’s $Ra1$ (a **cross-type** correspondence); cross-type roles: our $Bd7$ as their $Rf8$, our $Rb8$ as their $Pc6$; transferred move c6d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#556` (ply 45) [structural distance $d=66$, QAP obj=27]: the master played $Bxb5$; under μ our $Pc3$ plays the role of that game’s $Bc3$ (a **cross-type** correspondence); cross-type roles: our $Nf6$ as their $Rb8$, our $Bd7$ as their $Rd7$; transferred move c6d5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to cxd5 (summed QAP objective 120, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



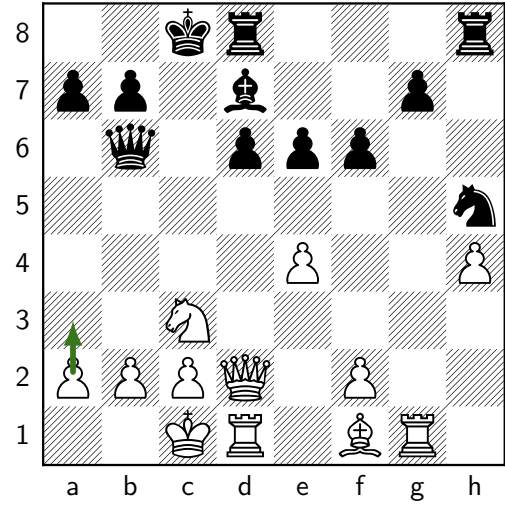
(a)



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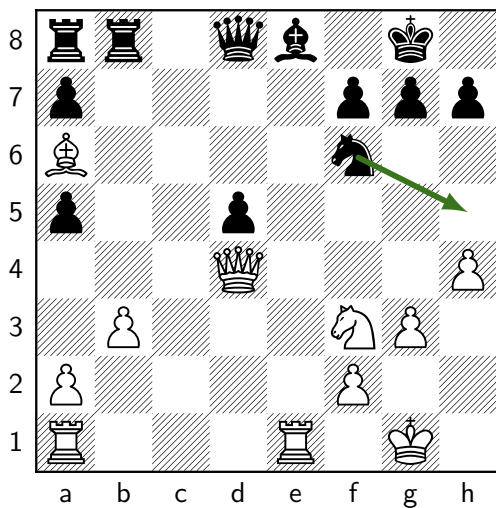


(c)

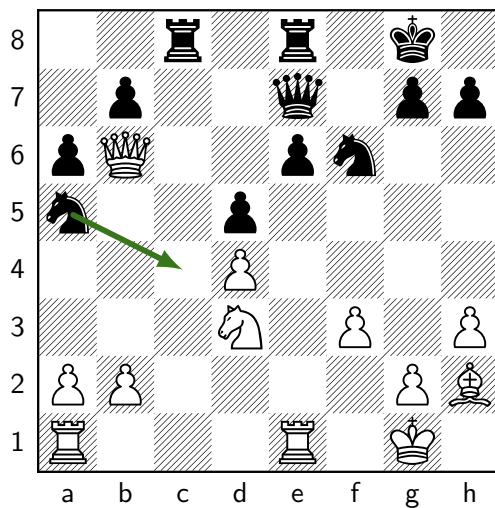


(d)

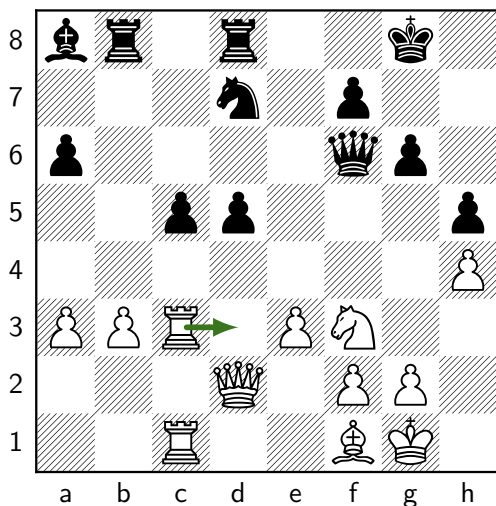
Figure 14: **Move 20, Black: Be8.** (a) Our position; Melanie plays Be8 (d7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#285` (ply 47) [structural distance $d=69$, QAP obj=65]: the master played Be6; under μ our Bd2 plays the role of that game’s Bf4; cross-type roles: our Nf6 as their Qd7, our Qd8 as their Nf6; transferred move d7e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2548` (ply 30) [structural distance $d=71$, QAP obj=64]: the master played Nb4; under μ our Bd2 plays the role of that game’s Nd3 (a **cross-type** correspondence); cross-type roles: our Nf6 as their Rd1, our Pf7 as their Rh1; transferred move d7a4. (d) Retrieved from `ccr1_top_p0000#335` (ply 32) [structural distance $d=60$, QAP obj=60]: the master played a3; under μ our Pa4 plays the role of that game’s Pa2; cross-type roles: our ba6 as their Rd1, our Nf6 as their Bf1; transferred move a5a4. Synthesis: 1 retrieved analog transferred to Be8 (summed QAP objective 65, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



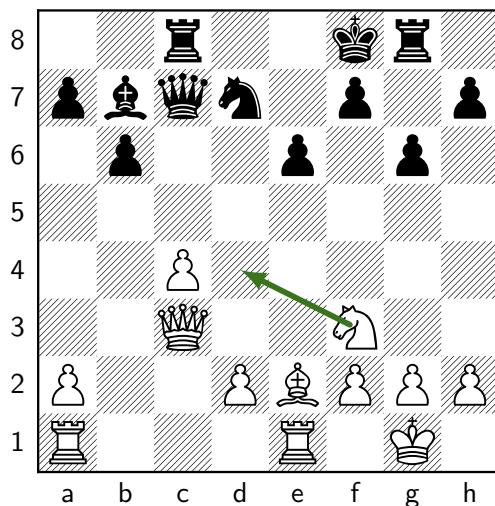
(a)



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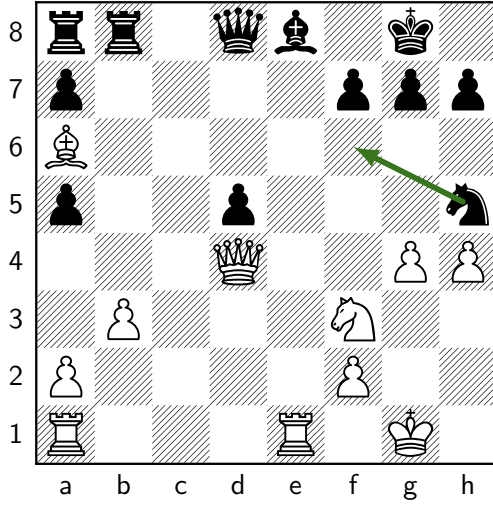


(c)

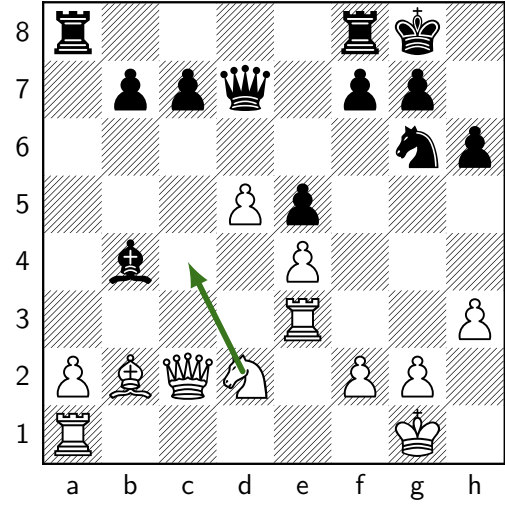


(d)

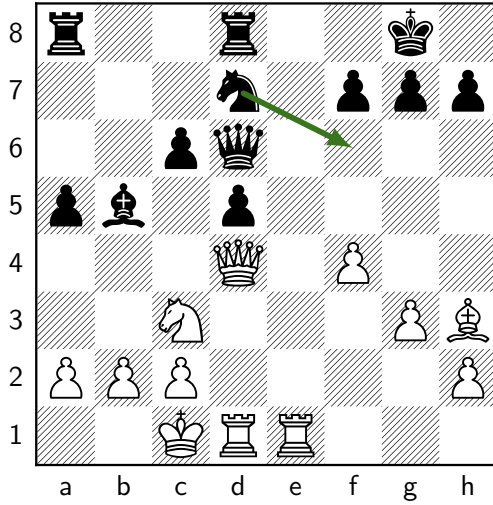
Figure 15: **Move 21, Black: Nh5**. (a) Our position; Melanie plays Nh5 (f6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2361` (ply 53) [structural distance $d=76$, QAP obj=51]: the master played Nc4; under μ our Nf3 plays the role of that game’s Na4; cross-type roles: our pg3 as their Rc8, our Be8 as their Re8; transferred move f6h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2930` (ply 46) [structural distance $d=72$, QAP obj=44]: the master played Rd3; under μ our Nf3 plays the role of that game’s Rc3 (a **cross-type** correspondence); cross-type roles: our Be8 as their Rc1, our Pa5 as their Bf1; transferred move f6h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3129` (ply 30) [structural distance $d=72$, QAP obj=25]: the master played Nd4; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Ph7 as their Re1, our Rb8 as their Pa2; transferred move f6h5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nh5 (summed QAP objective 120, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



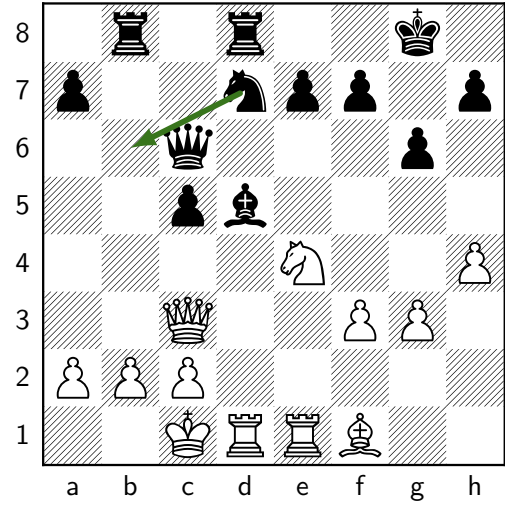
(a)



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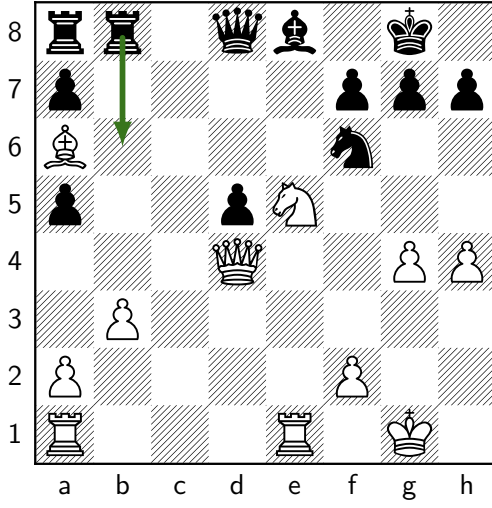


(c)

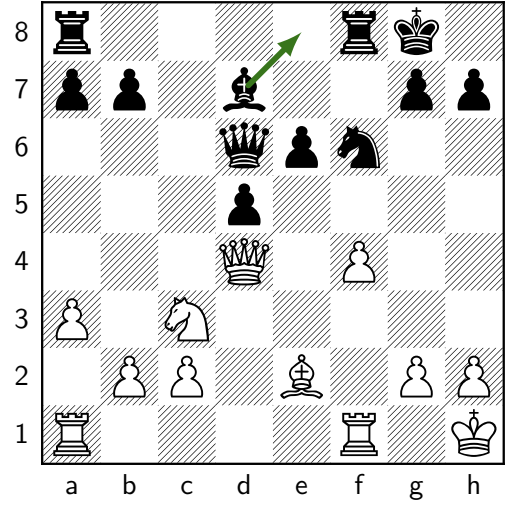


(d)

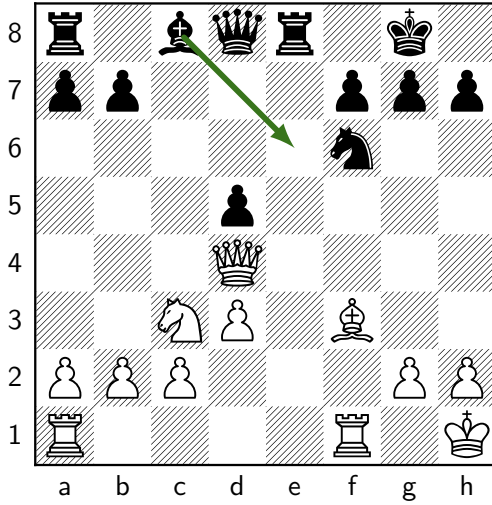
Figure 16: **Move 22, Black: Nf6.** (a) Our position; Melanie plays Nf6 (h5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1953` (ply 40) [structural distance $d=76$, QAP obj=28]: the master played Nc4; under μ our Nh4 plays the role of that game’s Nd2; cross-type roles: our pf2 as their Ra1, our re1 as their Qc2; transferred move h5f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3071` (ply 47) [structural distance $d=80$, QAP obj=49]: the master played Nf6; under μ our Pf2 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Be8 as their Rd8, our re1 as their Qd6; transferred move f7f6. (d) Retrieved from `ccrl_top_p0003#1609` (ply 49) [structural distance $d=75$, QAP obj=34]: the master played Nb6; under μ our Qd1 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Pa7, our Be8 as their Pe7; transferred move d8b6. Synthesis: 1 retrieved analog transferred to Nf6 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



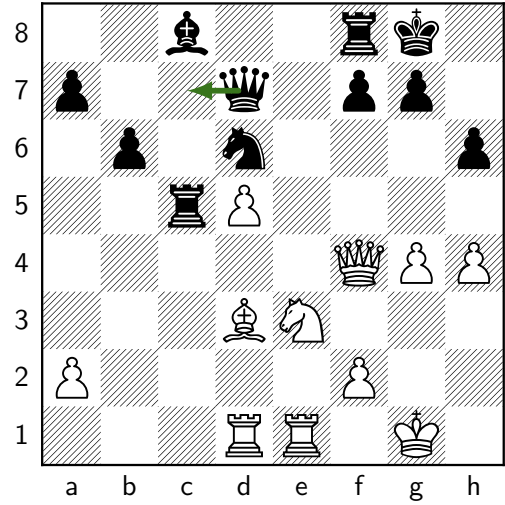
(a)



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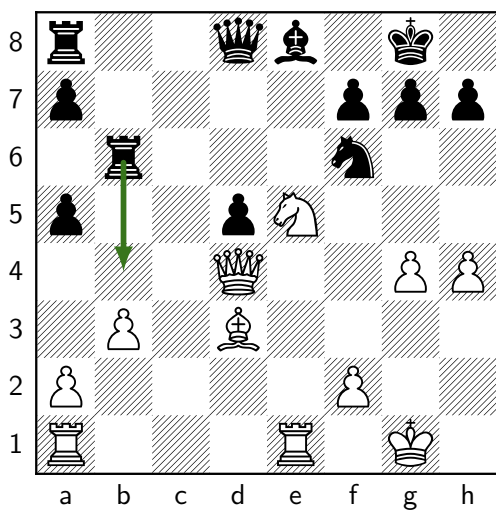


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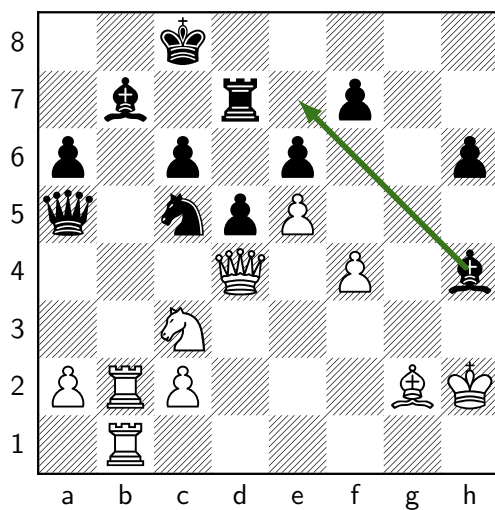


(d)

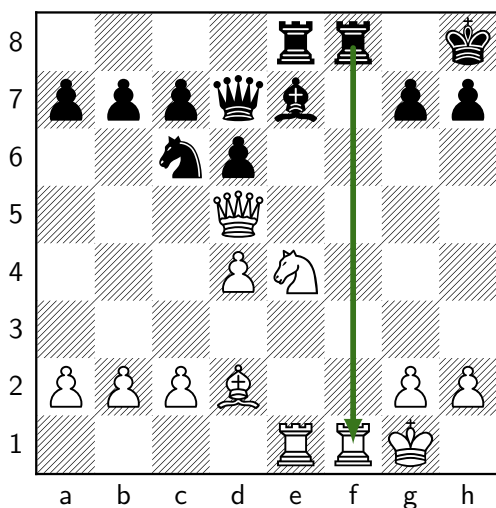
Figure 17: **Move 23, Black: Rb6**. (a) Our position; Melanie plays Rb6 (b8b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#555` (ply 41) [structural distance $d=47$, QAP obj=65]: the master played Be8; under μ our Rb1 plays the role of that game’s Bd2 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Rf8, our Nf6 as their Qd6; transferred move b8b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2893` (ply 33) [structural distance $d=58$, QAP obj=64]: the master played Be6; under μ our Rb1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Nf6 as their Re8, our Pd5 as their Nf6; transferred move b8c8. (d) Retrieved from `ccr1_top_p0001#1360` (ply 47) [structural distance $d=75$, QAP obj=47]: the master played Qc7; under μ our Nf3 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Bc8, our Rb8 as their Pb6; transferred move f6h5. Synthesis: 1 retrieved analog transferred to Rb6 (summed QAP objective 65, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



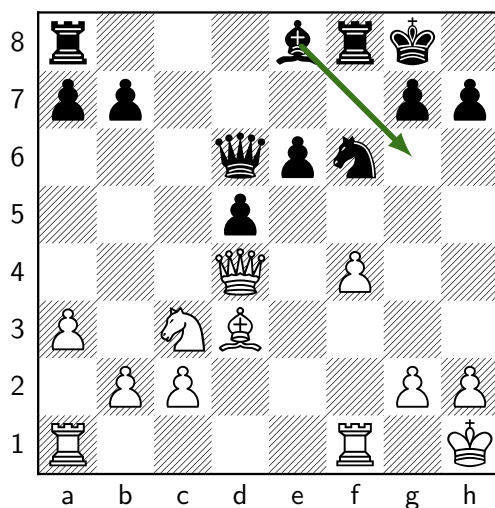
(a)



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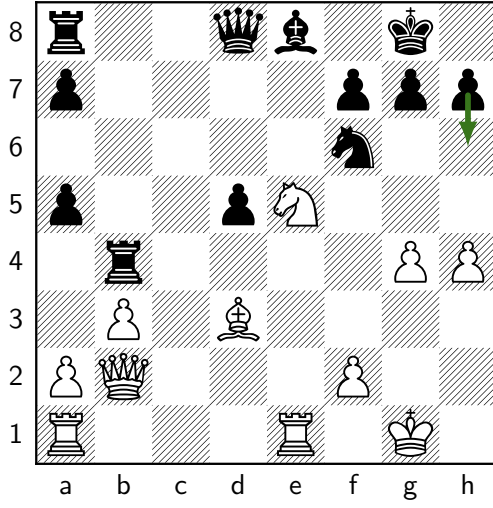


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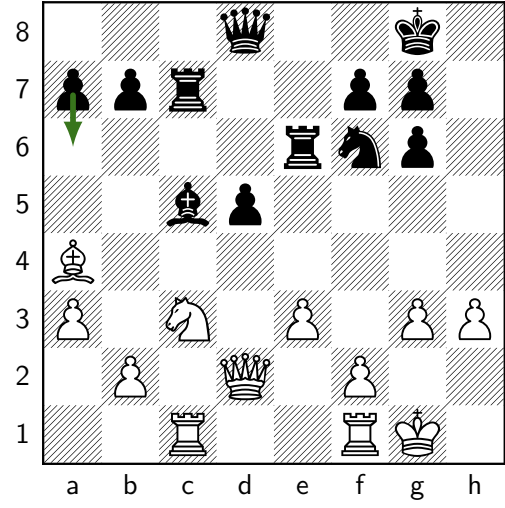


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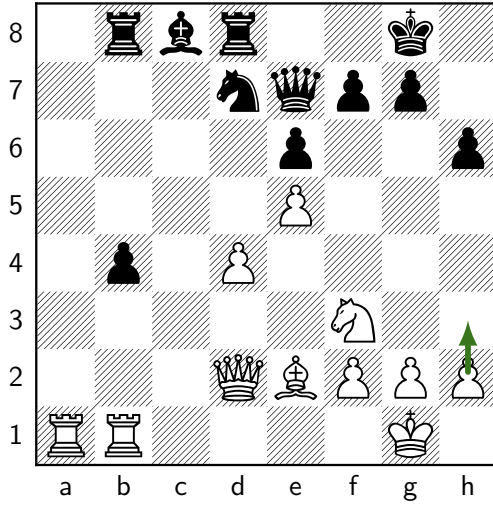
Figure 18: **Move 24, Black: Rb4**. (a) Our position; Melanie plays Rb4 (b6b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1280` (ply 47) [structural distance $d=92$, QAP obj=68]: the master played Be7; under μ our Be1 plays the role of that game’s Bh5; cross-type roles: our Ph7 as their Bb7, our Qd8 as their Rd7; transferred move e8c6. (c) Retrieved from `ccr1_top_p0000#3831` (ply 35) [structural distance $d=85$, QAP obj=66]: the master played Rxf1+; under μ our Nf3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Be7, our Rb6 as their Nc6; transferred move f6e4. (d) Retrieved from `ccr1_top_p0003#555` (ply 43) [structural distance $d=94$, QAP obj=60]: the master played Bg6; under μ our Pa4 plays the role of that game’s Be1 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Ra8, our Pf7 as their Rf8; transferred move a5a4. Synthesis: no retrieved analog transferred to Rb4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



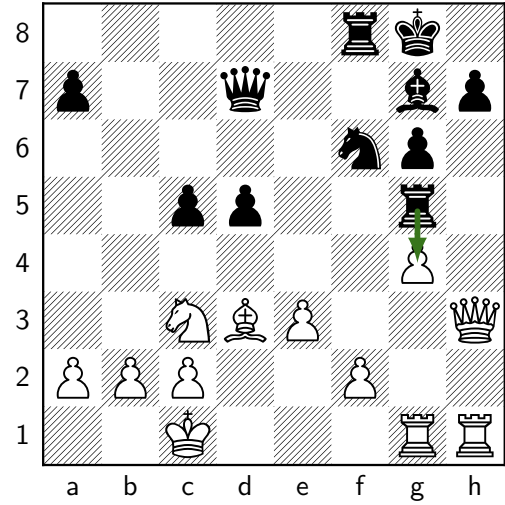
(a)



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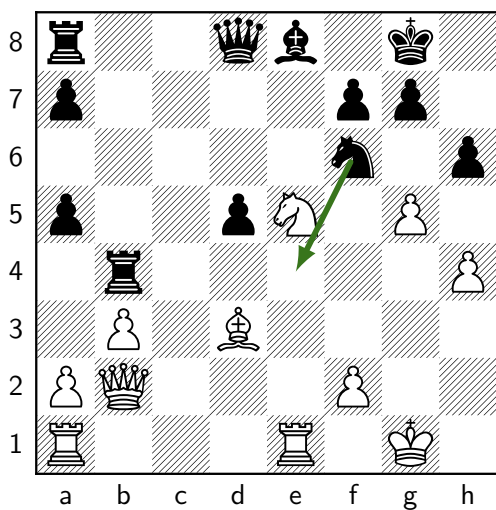


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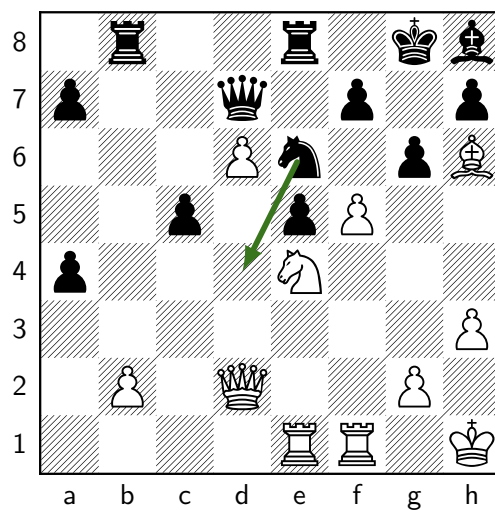


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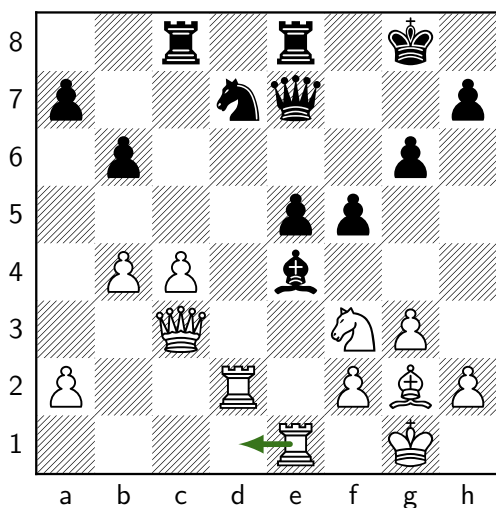
Figure 19: **Move 25, Black: h6**. (a) Our position; Melanie plays h6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1866` (ply 45) [structural distance $d=89$, QAP obj=39]: the master played a6; under μ our $Ph2$ plays the role of that game’s $Pa2$; cross-type roles: our $Qd8$ as their $Pb7$, our $Pf7$ as their $Rc7$; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3221` (ply 38) [structural distance $d=85$, QAP obj=35]: the master played h3; under μ our $Ph2$ plays the role of that game’s $Ph2$; cross-type roles: our $Pa5$ as their $Ra1$, our $Rb4$ as their $Qd2$; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2450` (ply 41) [structural distance $d=75$, QAP obj=77]: the master played Rxc4; under μ our $Rb5$ plays the role of that game’s $Rg4$; cross-type roles: our $Pd5$ as their $Rf8$, our $Rb4$ as their $Qd7$; transferred move b4e4. Synthesis: 2 retrieved analogs transferred to h6 (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



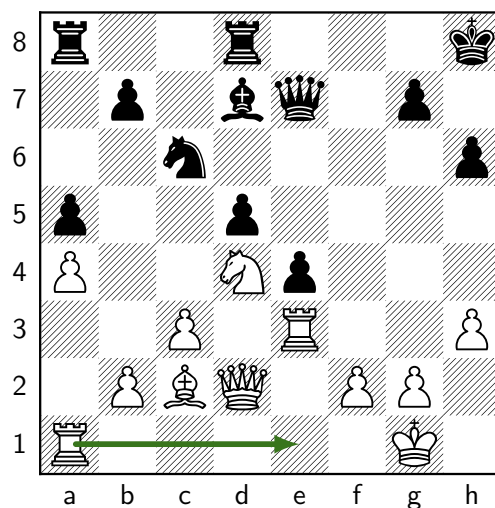
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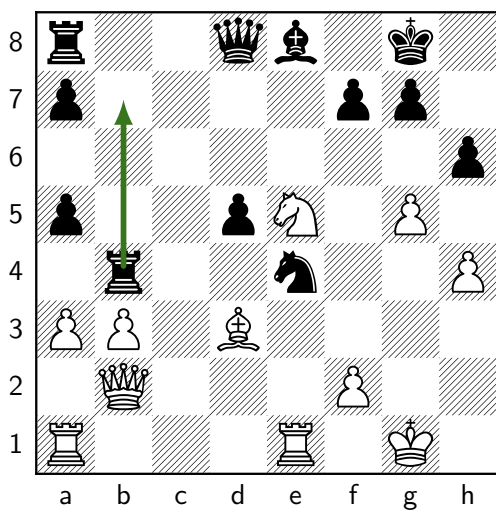


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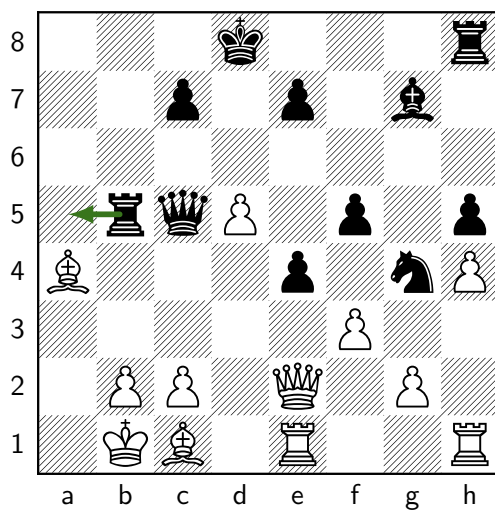


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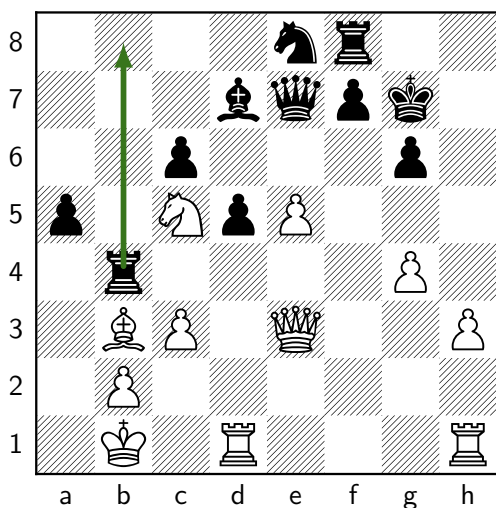
Figure 20: **Move 26, Black: Ne4**. (a) Our position; Melanie plays Ne4 (f6e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0002#2737` (ply 55) [structural distance $d=76$, QAP obj=81]: the master played Nd4; under μ our Nf3 plays the role of that game’s Ne3; cross-type roles: our Qd8 as their Rb8, our ne5 as their Bh8; transferred move f6e4 (**endorses this move**). (c) Retrieved from `cctl_top_p0002#495` (ply 42) [structural distance $d=81$, QAP obj=54]: the master played Red1; under μ our Ph3 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pa2, our Be8 as their Rd2; transferred move h6h5. (d) Retrieved from `cctl_top_p0000#2516` (ply 48) [structural distance $d=75$, QAP obj=43]: the master played Rae1; under μ our Ra1 plays the role of that game’s Ra1; cross-type roles: our Pa5 as their Bc2, our ra1 as their Qd2; transferred move a8b8. Synthesis: 1 retrieved analog transferred to Ne4 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



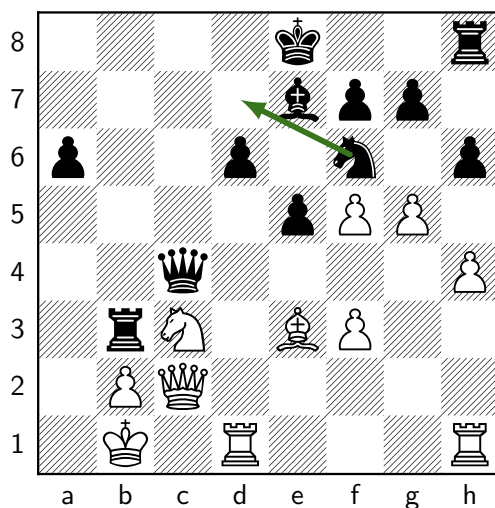
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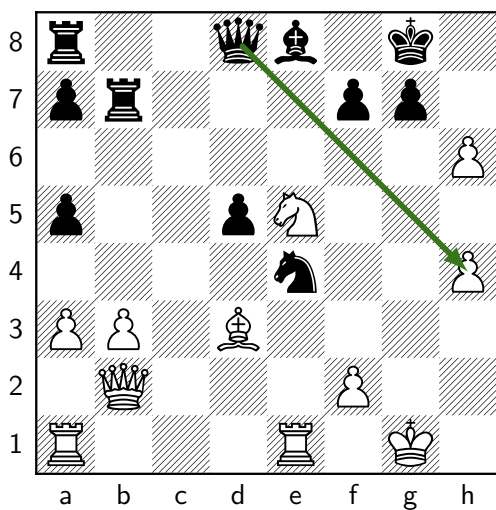


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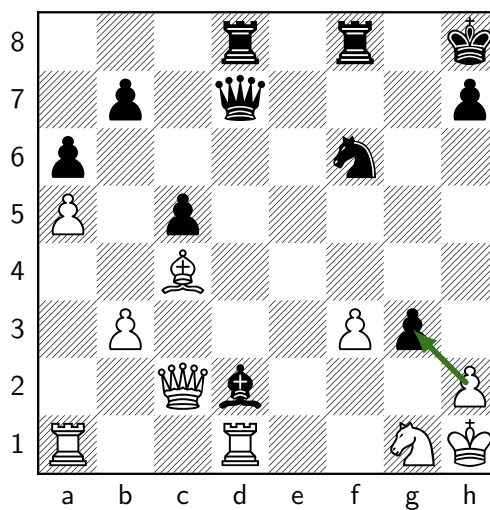


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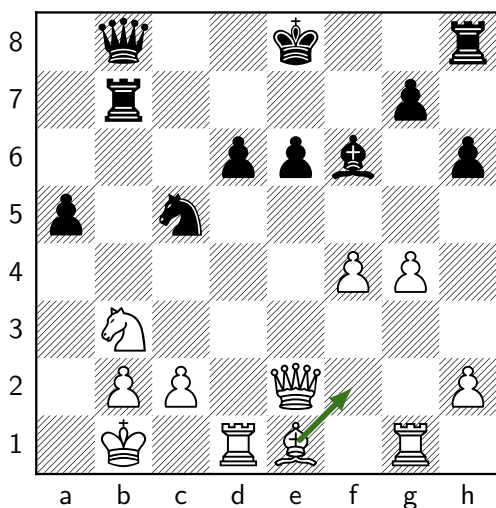
Figure 21: **Move 27, Black: Rb7**. (a) Our position; Melanie plays Rb7 (b4b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1850` (ply 51) [structural distance $d=89$, QAP obj=72]: the master played Ra5; under μ our Rb5 plays the role of that game’s Rb4; cross-type roles: our Qd8 as their Rh8, our ne5 as their Bg7; transferred move b4b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#270` (ply 47) [structural distance $d=70$, QAP obj=41]: the master played Rb8; under μ our Rb5 plays the role of that game’s Rb5; cross-type roles: our Pd5 as their Ne8, our Pf7 as their Rf8; transferred move b4b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3851` (ply 43) [structural distance $d=79$, QAP obj=68]: the master played Nd7; under μ our Ne5 plays the role of that game’s Nf3; cross-type roles: our Pf7 as their Be7, our Be8 as their Ph6; transferred move e4d6. Synthesis: 2 retrieved analogs transferred to Rb7 (summed QAP objective 113, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



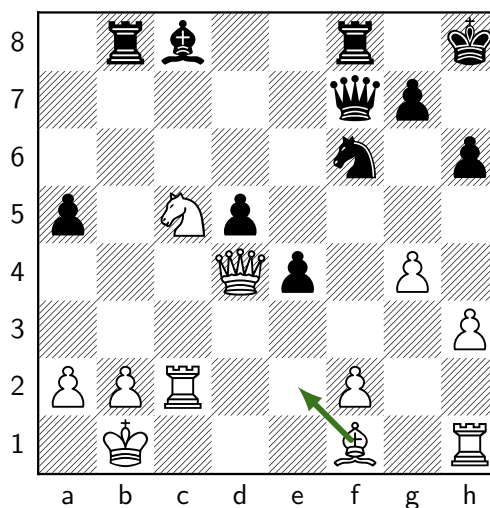
(a)



(b)

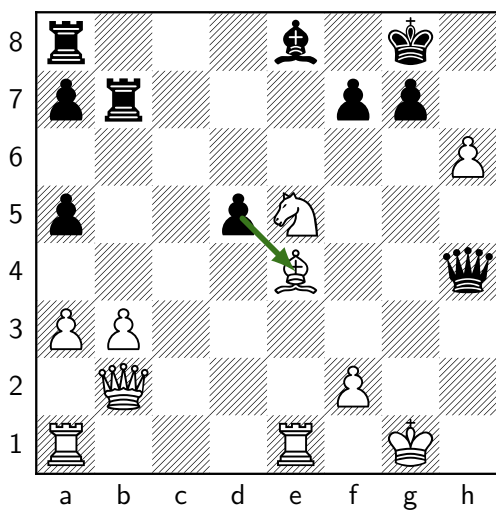


(c)

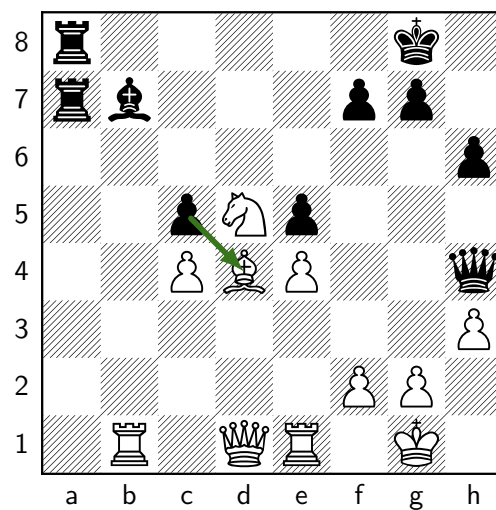


(d)

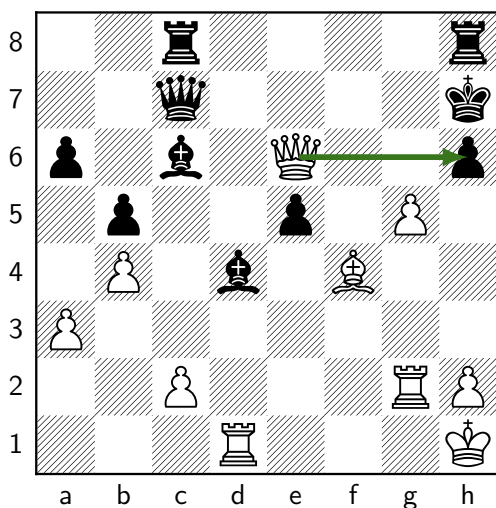
Figure 22: **Move 28, Black: Qxh4.** (a) Our position; Melanie plays Qxh4 (d8h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3765` (ply 62) [structural distance $d=86$, QAP obj=59]: the master played **hxg3**; under μ our *Pg2* plays the role of that game’s *Ph2*; cross-type roles: our *qb2* as their *Ra1*, our *bd3* as their *Rd1*; transferred move **g7h6**. (c) Retrieved from `ccr1_top_p0003#1712` (ply 46) [structural distance $d=72$, QAP obj=41]: the master played **Bf2**; under μ our *Pa2* plays the role of that game’s *Be1* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Rg1*, our *Rb7* as their *Qe2*; transferred move **a7a6**. (d) Retrieved from `ccr1_top_p0000#494` (ply 54) [structural distance $d=85$, QAP obj=38]: the master played **Be2**; under μ our *Qd1* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Be8* as their *Rh1*, our *qb2* as their *Rc2*; transferred move **d8c7**. Synthesis: no retrieved analog transferred to Qxh4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



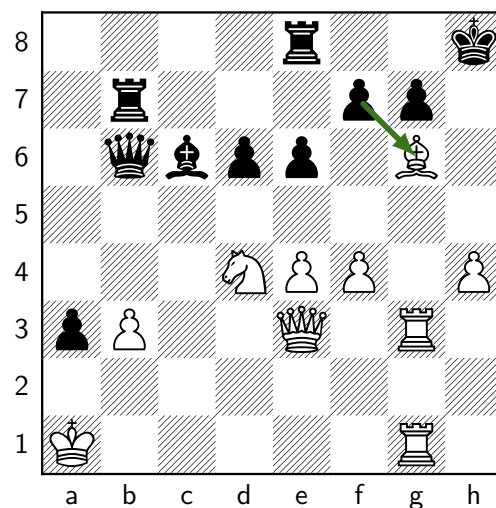
(a)



(b)

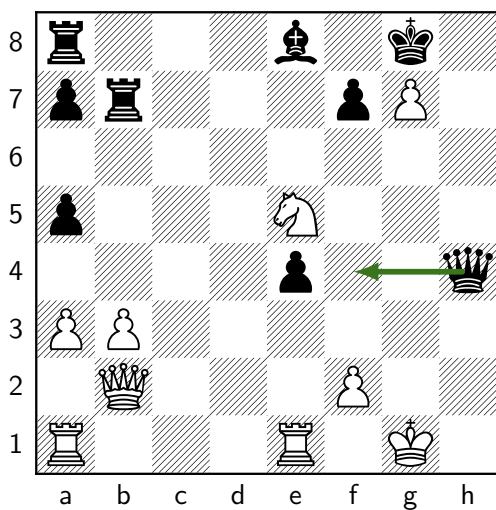


(c)

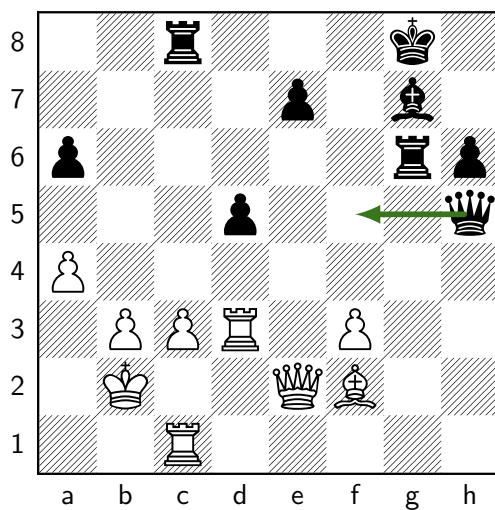


(d)

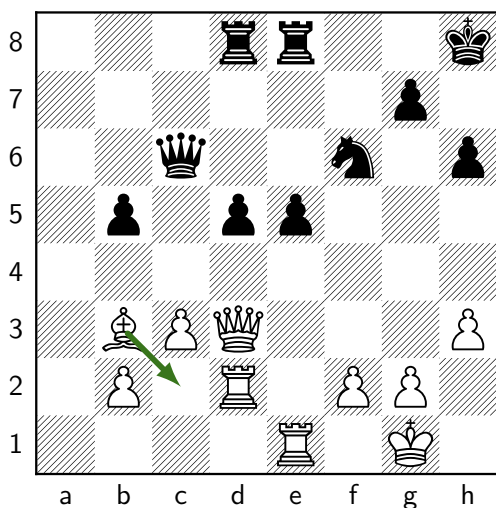
Figure 23: **Move 29, Black: dxe4**. (a) Our position; Melanie plays dxe4 (d5e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3835 (ply 73) [structural distance $d=96$, QAP obj=103]: the master played cxd4; under μ our $Pd4$ plays the role of that game’s $Pc4$; cross-type roles: our $qb2$ as their $Ra7$, our $ph6$ as their $Bb7$; transferred move d5e4 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1943 (ply 70) [structural distance $d=74$, QAP obj=81]: the master played Qxh6+; under μ our $Pd4$ plays the role of that game’s $Qe6$ (a **cross-type** correspondence); cross-type roles: our $ne5$ as their $Rd1$, our $re1$ as their $Pc2$; transferred move d5e4 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3688 (ply 67) [structural distance $d=90$, QAP obj=59]: the master played fxg6; under μ our $Pd4$ plays the role of that game’s $Pf2$; cross-type roles: our $ne5$ as their $Rb7$, our $be4$ as their $Qb6$; transferred move d5e4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to dxe4 (summed QAP objective 342, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



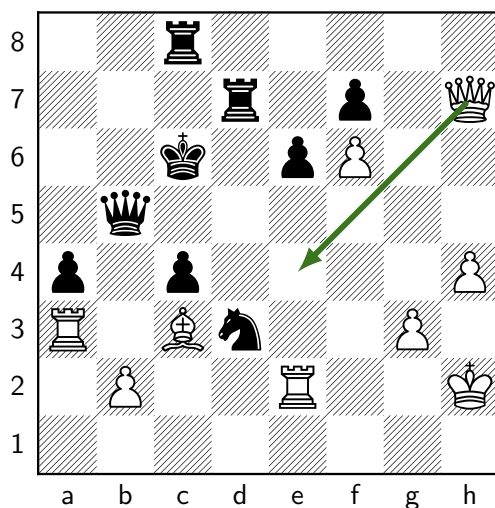
(a)



(b)

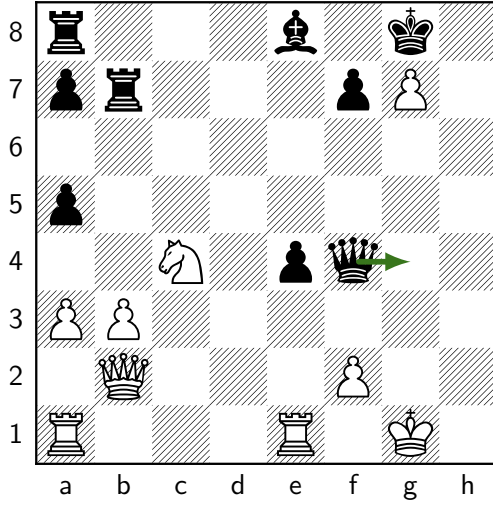


(c)

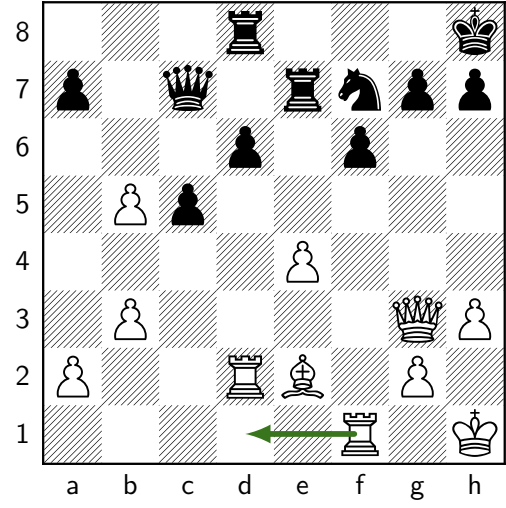


(d)

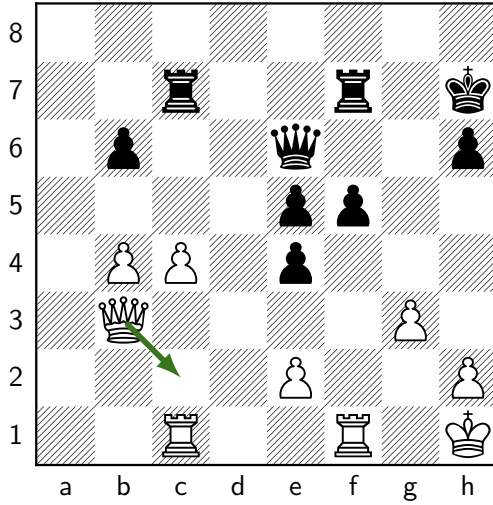
Figure 24: **Move 30, Black: Qf4 (h4f4).** (a) Our position; Melanie plays Qf4 (h4f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2575` (ply 69) [structural distance $d=64$, QAP obj=60]: the master played Qf5; under μ our Qh5 plays the role of that game’s Qh4; cross-type roles: our Rb7 as their Bg7, our Be8 as their pa4; transferred move h4f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3795` (ply 70) [structural distance $d=82$, QAP obj=65]: the master played Bc2; under μ our Qh5 plays the role of that game’s Bb3 (a **cross-type** correspondence); cross-type roles: our Be8 as their Pb2, our ne5 as their Ph3; transferred move h4e7. (d) Retrieved from `ccr1_top_p0002#1426` (ply 72) [structural distance $d=81$, QAP obj=64]: the master played Qe4+; under μ our Qh5 plays the role of that game’s Qh7; cross-type roles: our Kg8 as their Ra3, our ne5 as their qb5; transferred move h4e7. Synthesis: 1 retrieved analog transferred to Qf4 (summed QAP objective 60, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



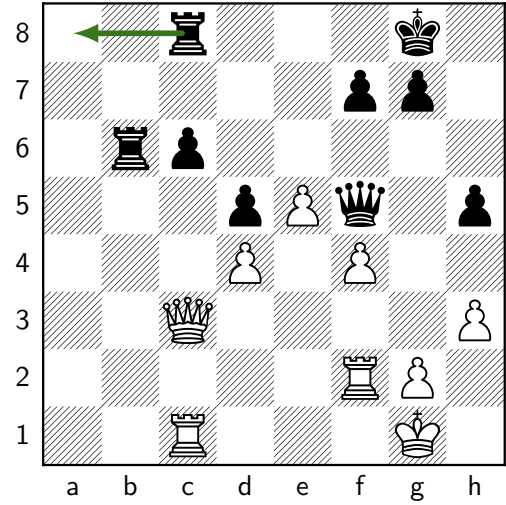
(a)



(b)

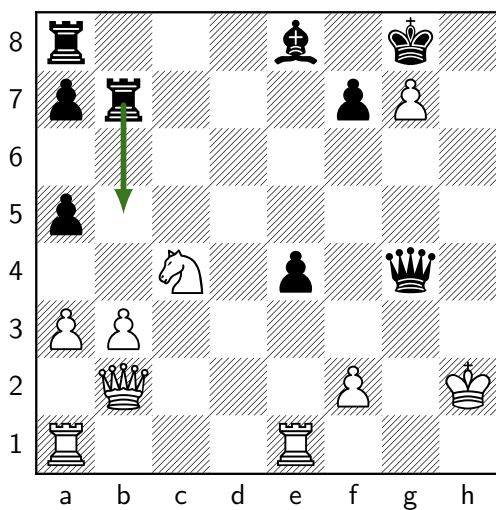


(c)

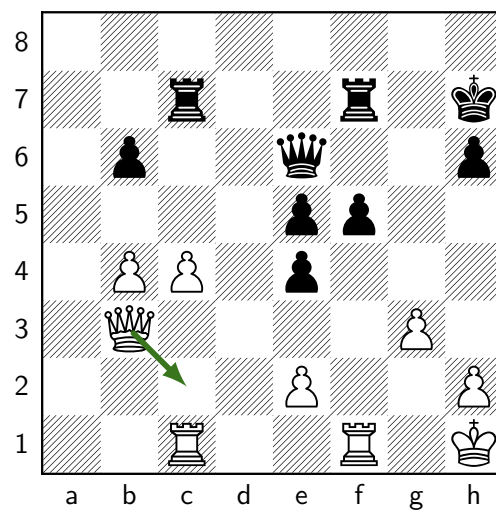


(d)

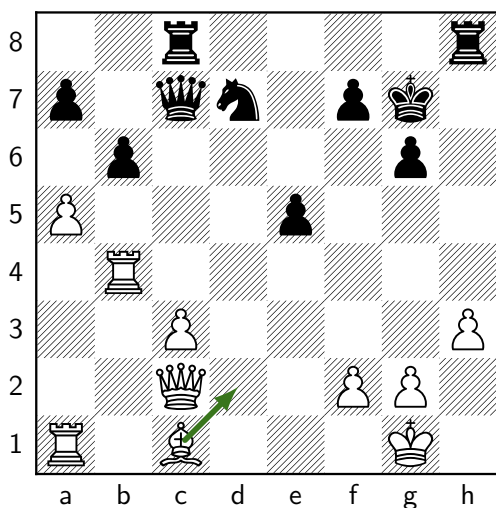
Figure 25: **Move 31, Black: Qg4+.** (a) Our position; Melanie plays Qg4+ (f4g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1470` (ply 56) [structural distance $d=76$, QAP obj=74]: the master played Rfd1; under μ our Rb2 plays the role of that game’s Rf1; cross-type roles: our Pa5 as their Be2, our Ra8 as their Pg2; transferred move b7b8. (c) Retrieved from `ccrl_top_p0002#416` (ply 66) [structural distance $d=72$, QAP obj=58]: the master played Qc2; under μ our Qf5 plays the role of that game’s Qb3; cross-type roles: our Pa7 as their Rc1, our Ra8 as their Pe2; transferred move f4c7. (d) Retrieved from `ccrl_top_p0002#1273` (ply 57) [structural distance $d=79$, QAP obj=50]: the master played Ra8; under μ our Be1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Rb7 as their Pg7, our Qf4 as their Rb6; transferred move e8d7. Synthesis: no retrieved analog transferred to Qg4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



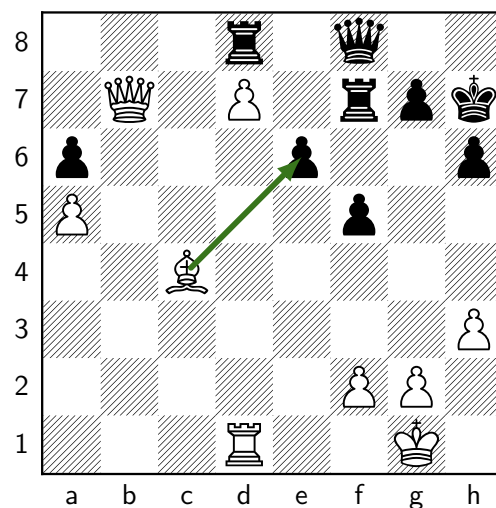
(a)



(b)

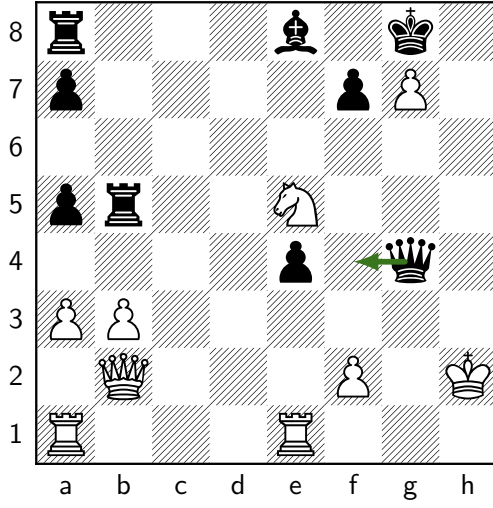


(c)

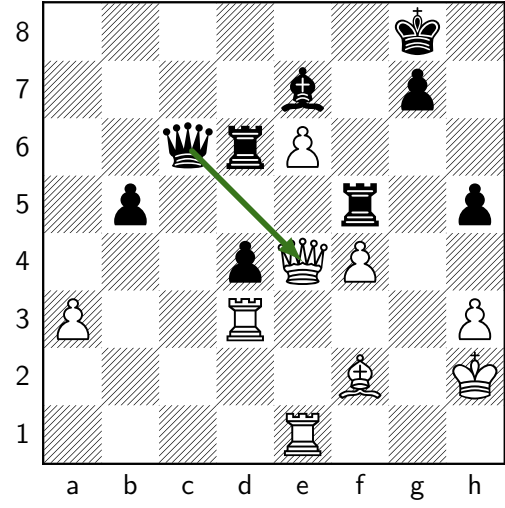


(d)

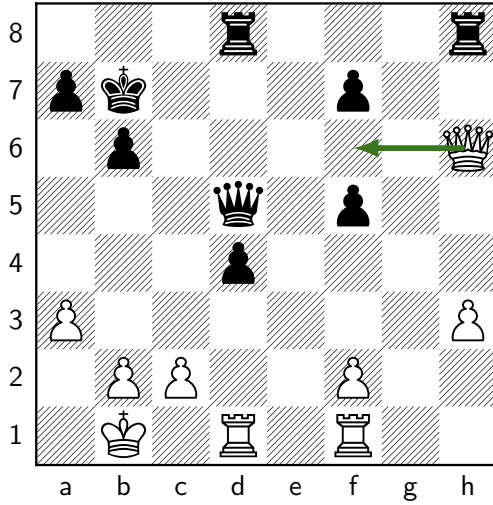
Figure 26: **Move 32, Black: Rb5**. (a) Our position; Melanie plays Rb5 (b7b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#416` (ply 66) [structural distance $d=76$, QAP obj=59]: the master played Qc2; under μ our Qg5 plays the role of that game’s Qb3; cross-type roles: our Pa7 as their Rc1, our Ra8 as their Pb4; transferred move g4c8. (c) Retrieved from `ccr1_top_p0001#2204` (ply 46) [structural distance $d=84$, QAP obj=49]: the master played Bd2; under μ our Be1 plays the role of that game’s Bc1; cross-type roles: our Rb7 as their Qc2, our Qg4 as their Rb4; transferred move e8b5. (d) Retrieved from `ccr1_top_p0000#944` (ply 74) [structural distance $d=83$, QAP obj=49]: the master played Bxe6; under μ our Rb2 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our Qg4 as their Rd1, our Ra8 as their Pf2; transferred move b7b3. Synthesis: no retrieved analog transferred to Rb5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



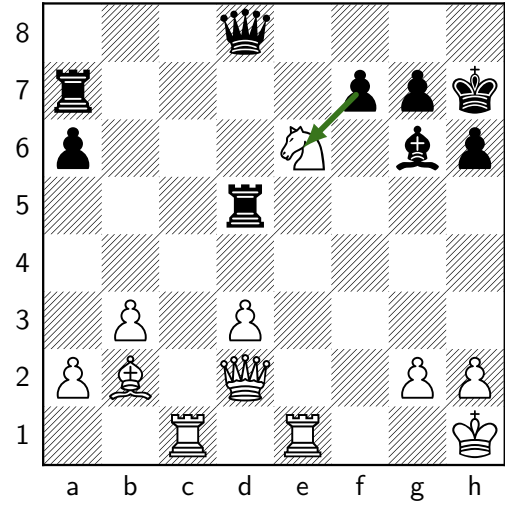
(a)



(b)

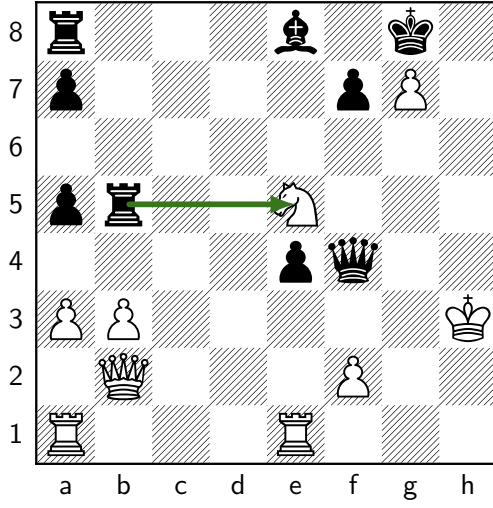


(c)

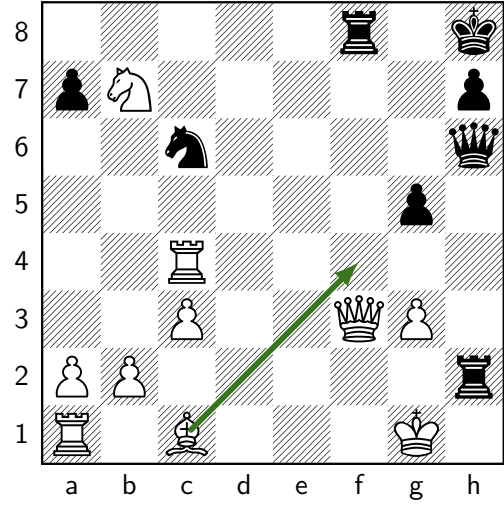


(d)

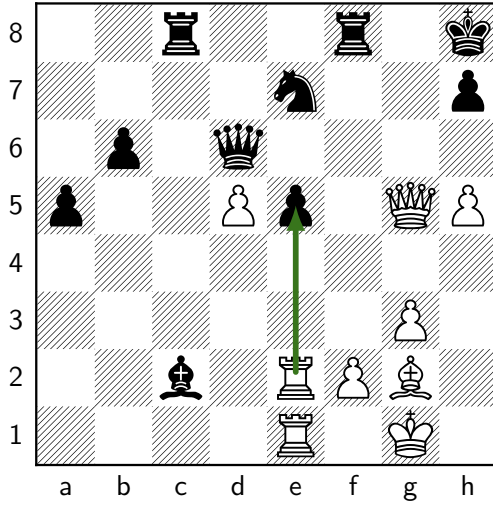
Figure 27: **Move 33, Black: Qf4+.** (a) Our position; Melanie plays Qf4+ (g4f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2963` (ply 69) [structural distance $d=70$, QAP obj=43]: the master played Qxe4; under μ our Qg5 plays the role of that game’s Qc3; cross-type roles: our pa3 as their Be7, our Be8 as their Pg7; transferred move g4f4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2380` (ply 50) [structural distance $d=83$, QAP obj=37]: the master played Qf6; under μ our Qg5 plays the role of that game’s Qh6; cross-type roles: our Be8 as their Rf1, our re1 as their pf5; transferred move g4f4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#836` (ply 51) [structural distance $d=82$, QAP obj=52]: the master played fxe6; under μ our Rb4 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Pa5 as their Bg6, our Be8 as their pb3; transferred move b5e5. Synthesis: 2 retrieved analogs transferred to Qf4+ (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



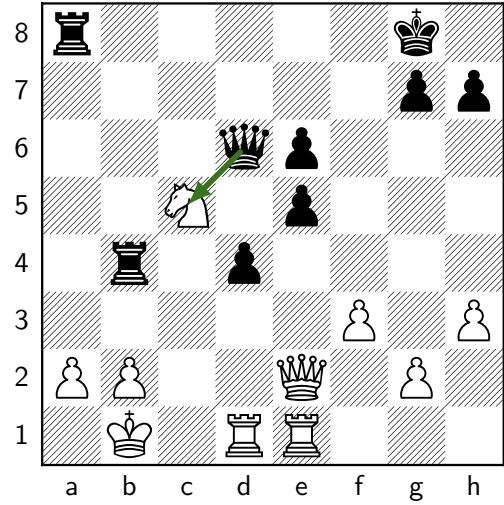
(a)



(b)

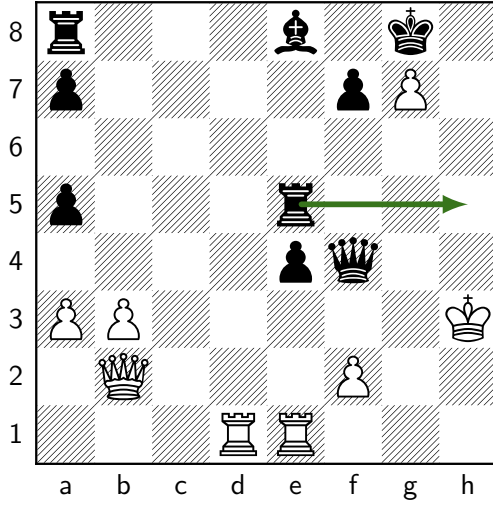


(c)

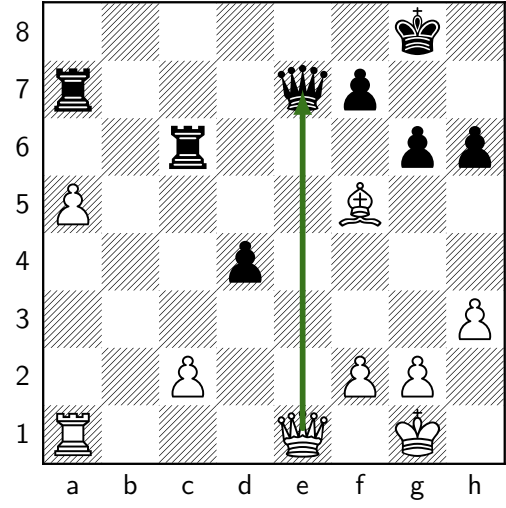


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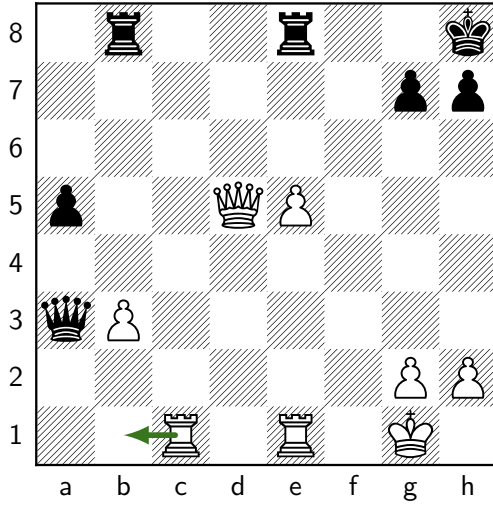
Figure 28: **Move 34, Black: Rxe5**. (a) Our position; Melanie plays Rxe5 (b5e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2211` (ply 60) [structural distance $d=54$, QAP obj=21]: the master played Bf4; under μ our Rb4 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Be8 as their Ra1, our pg7 as their rh2; transferred move b5e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1810` (ply 64) [structural distance $d=68$, QAP obj=67]: the master played Rxe5; under μ our Rb4 plays the role of that game’s Re2; cross-type roles: our Be8 as their Re1, our ne5 as their bc2; transferred move b5d5. (d) Retrieved from `ccr1_top_p0000#2227` (ply 51) [structural distance $d=81$, QAP obj=52]: the master played Qxc5; under μ our Qf5 plays the role of that game’s Qd3; cross-type roles: our Be8 as their Pg7; transferred move f4e5. Synthesis: 1 retrieved analog transferred to Rxe5 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



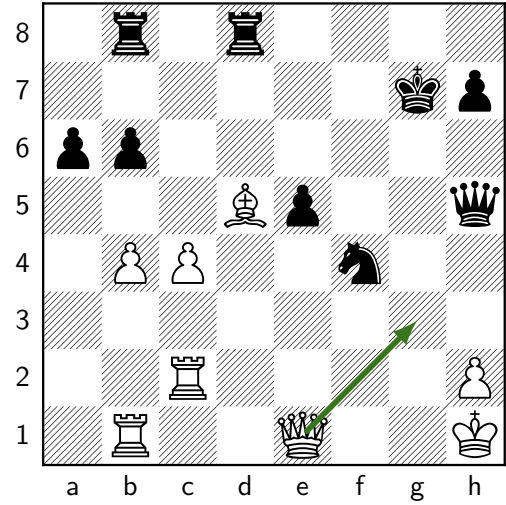
(a)



(b)

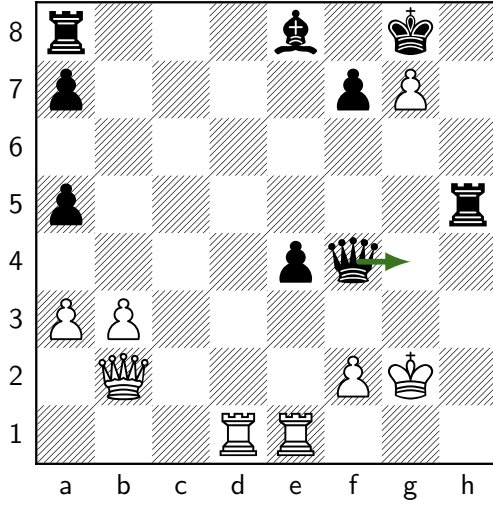


(c)

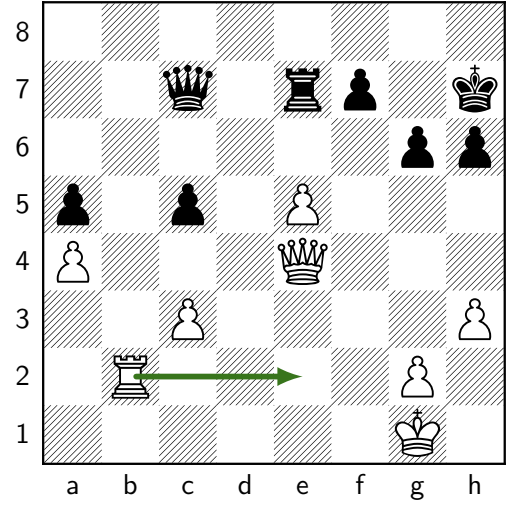


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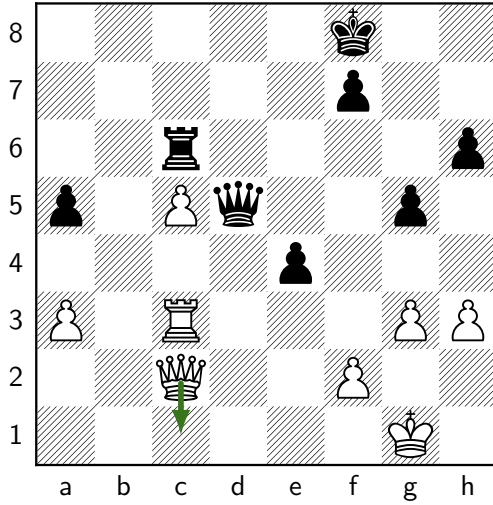
Figure 29: **Move 35, Black: Rh5+.** (a) Our position; Melanie plays Rh5+ (e5h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1855` (ply 70) [structural distance $d=77$, QAP obj=29]: the master played Qxe7; under μ our Re4 plays the role of that game’s Qe1 (a **cross-type** correspondence); cross-type roles: our Qf4 as their Ra1, our Be8 as their Pf2; transferred move e5h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2038` (ply 60) [structural distance $d=77$, QAP obj=47]: the master played Rb1; under μ our Re4 plays the role of that game’s Rc1; cross-type roles: our Be8 as their Re1, our Ra8 as their Ph2; transferred move e5e7. (d) Retrieved from `ccr1_top_p0002#1845` (ply 72) [structural distance $d=78$, QAP obj=47]: the master played Qg3+; under μ our Pf2 plays the role of that game’s Qe1 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Rb1, our Qf4 as their Rc2; transferred move f7f6. Synthesis: 1 retrieved analog transferred to Rh5+ (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



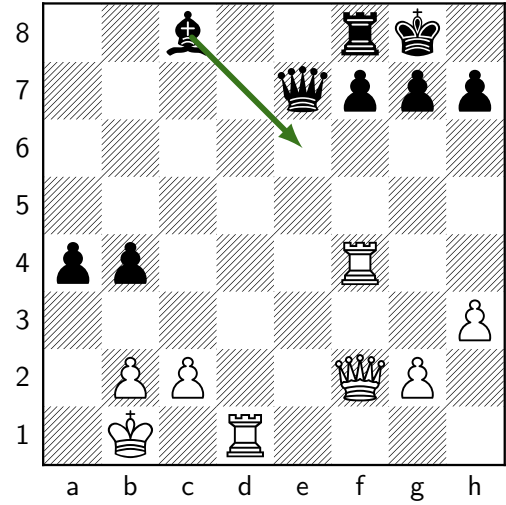
(a)



(b)

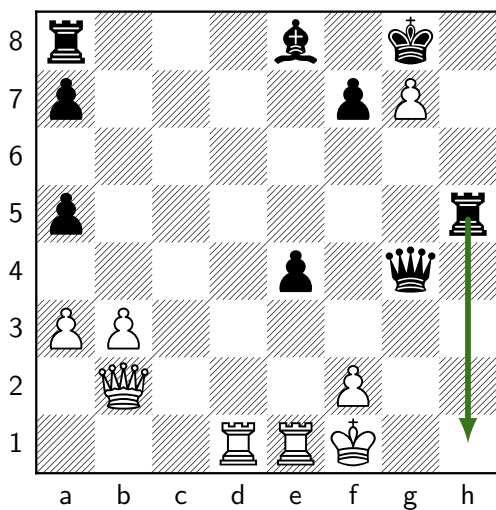


(c)

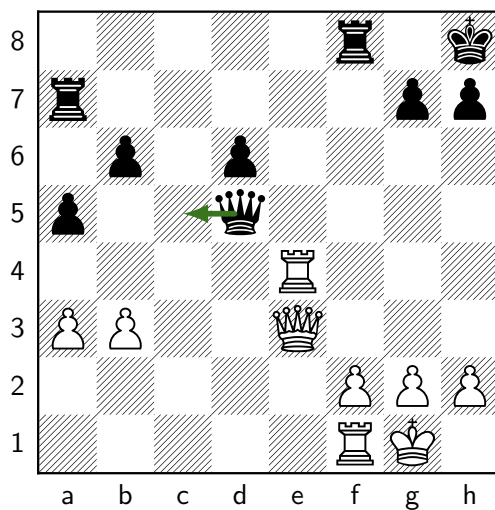


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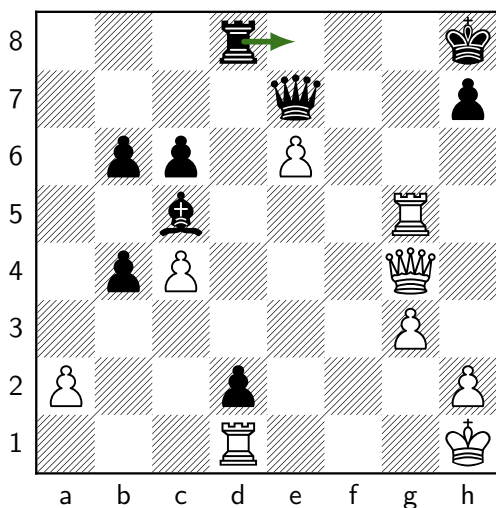
Figure 30: **Move 36, Black: Qg4+.** (a) Our position; Melanie plays Qg4+ (f4g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1719` (ply 64) [structural distance $d=54$, QAP obj=40]: the master played Re2; under μ our Be1 plays the role of that game’s Rb2 (a **cross-type** correspondence); cross-type roles: our Rh5 as their Ph3, our Ra8 as their Pa4; transferred move e8b5. (c) Retrieved from `ccrl_top_p0003#1520` (ply 76) [structural distance $d=43$, QAP obj=39]: the master played Qc1; under μ our Qf5 plays the role of that game’s Qc2; cross-type roles: our Be8 as their Pa3, our qb2 as their Rc3; transferred move f4h6. (d) Retrieved from `ccrl_top_p0001#3233` (ply 61) [structural distance $d=60$, QAP obj=36]: the master played Be6; under μ our Rh4 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Be8 as their Rf8, our Ra8 as their Ph7; transferred move h5h6. Synthesis: no retrieved analog transferred to Qg4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



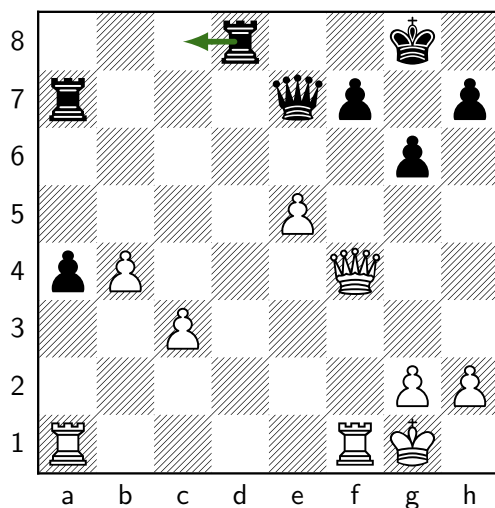
(a)



(b)



(c)



(d)

Figure 31: **Move 37, Black: Rh1#.** (a) Our position; Melanie plays Rh1# (h5h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1721` (ply 61) [structural distance $d=59$, QAP obj=33]: the master played Qc5; under μ our Qg5 plays the role of that game’s Qd4; cross-type roles: our Be8 as their Ra7, our Rh5 as their Pd6; transferred move g4e6. (c) Retrieved from `ccr1_top_p0001#1472` (ply 89) [structural distance $d=60$, QAP obj=26]: the master played Re8; under μ our Rh4 plays the role of that game’s Rd1; cross-type roles: our rd1 as their pe6, our Ra8 as their pa2; transferred move h5h4. (d) Retrieved from `ccr1_top_p0003#2270` (ply 65) [structural distance $d=53$, QAP obj=26]: the master played Rc8; under μ our Ra1 plays the role of that game’s Rd1; cross-type roles: our Pa5 as their Ra7, our Be8 as their Pg6; transferred move a8b8. Synthesis: no retrieved analog transferred to Rh1# – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.